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Fosler et al.

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(54) **DRILLING TOOL CHANGER APPARATUS**

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Related U.S. Application Data

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filed on Aug. 30, 2022, now Pat. No. 11,898,406,
which is a continuation of application No.
17/161,880, filed on Jan. 29, 2021, now Pat. No.
11,466,522.

(51) **Int. Cl.**
E21B 19/14 (2006.01)
E21B 19/20 (2006.01)

(52) **U.S. Cl.**
CPC **E21B 19/20** (2013.01); **E21B 19/146**
(2013.01)

(58) **Field of Classification Search**
CPC E21B 19/146; E21B 19/20
See application file for complete search history.

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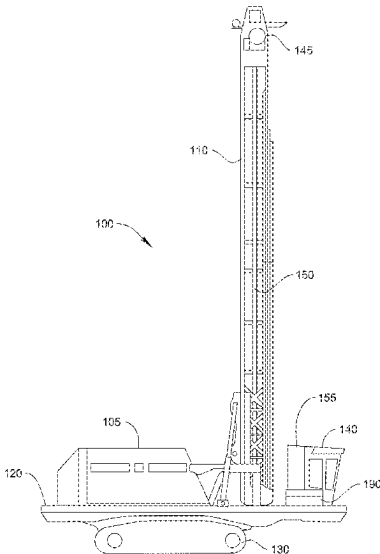
Primary Examiner — Matthew R Buck

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Harper & Bates LLP

(57) **ABSTRACT**

An apparatus for changing a drilling tool for a drilling rig is described, where the drilling rig comprises a drill tower supporting a drill pipe. The apparatus for changing a drilling tool has a drilling tool changer assembly moveable between a storage position and an exchange position, and the drilling tool changer assembly supports a rotatable carousel assembly. The rotatable carousel assembly has a plurality of bit adaptors for holding drilling tools and is selectively rotatable to bring a selected drilling tool into coaxial alignment with the drill pipe when the drilling tool changer assembly supporting the rotatable carousel assembly is moved into the exchange position. The apparatus includes mechanisms for aligning the selected drilling tool within a bit adaptor holding the selected drilling tool. The apparatus further includes mechanisms for aligning the drill pipe or a hammer with the selected drilling tool.

17 Claims, 40 Drawing Sheets



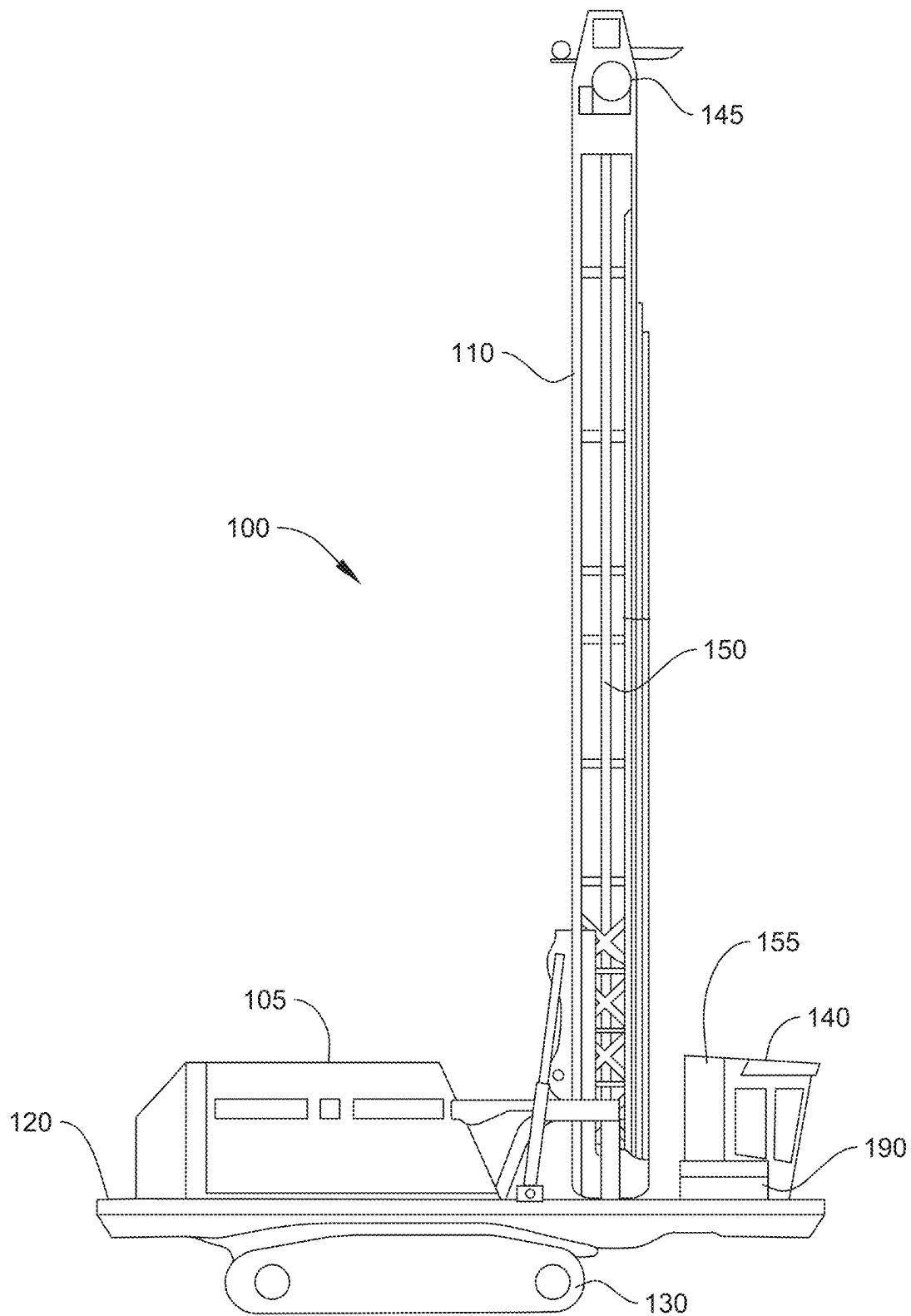


FIG. 1

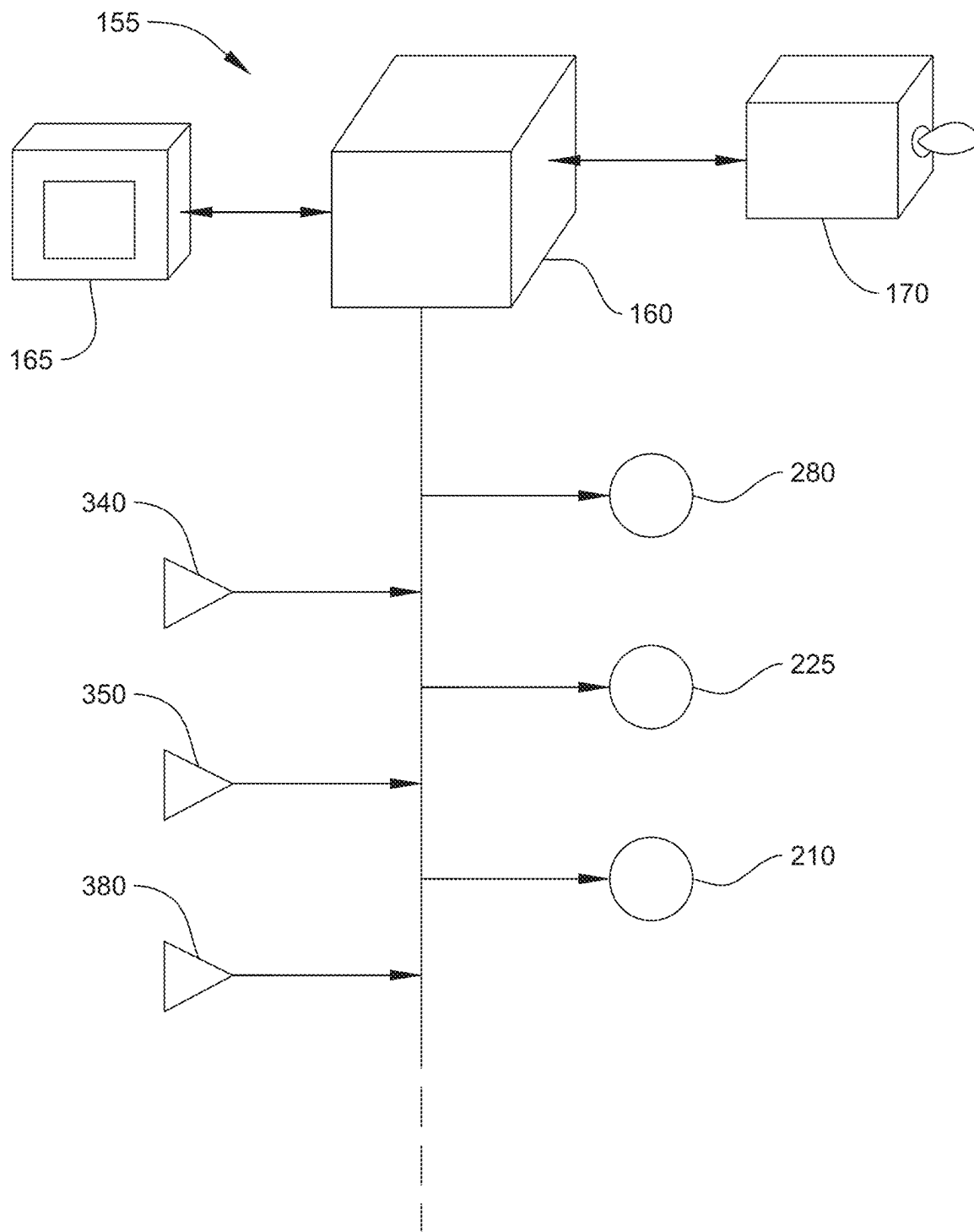


FIG. 2

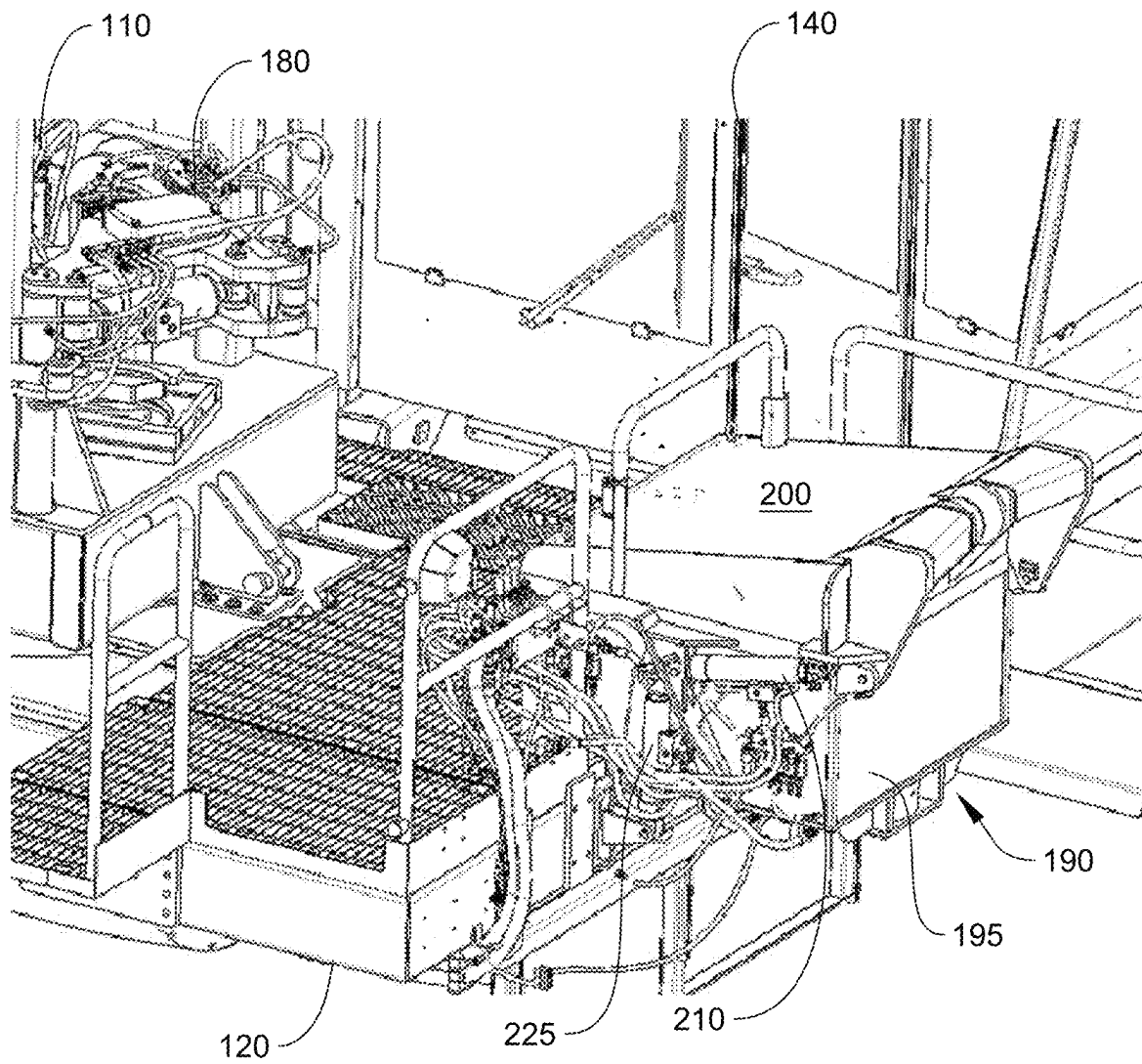


FIG. 3

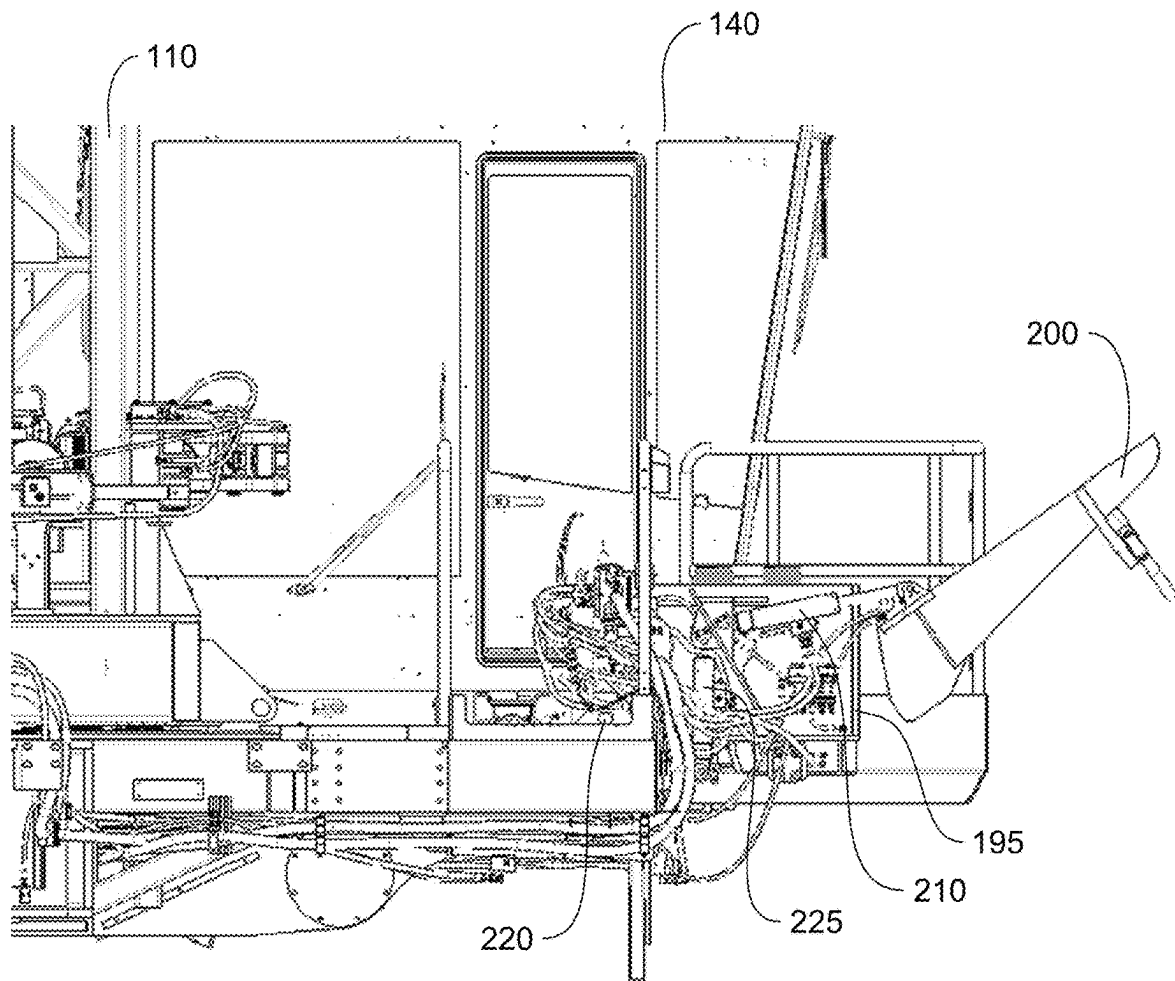


FIG. 4

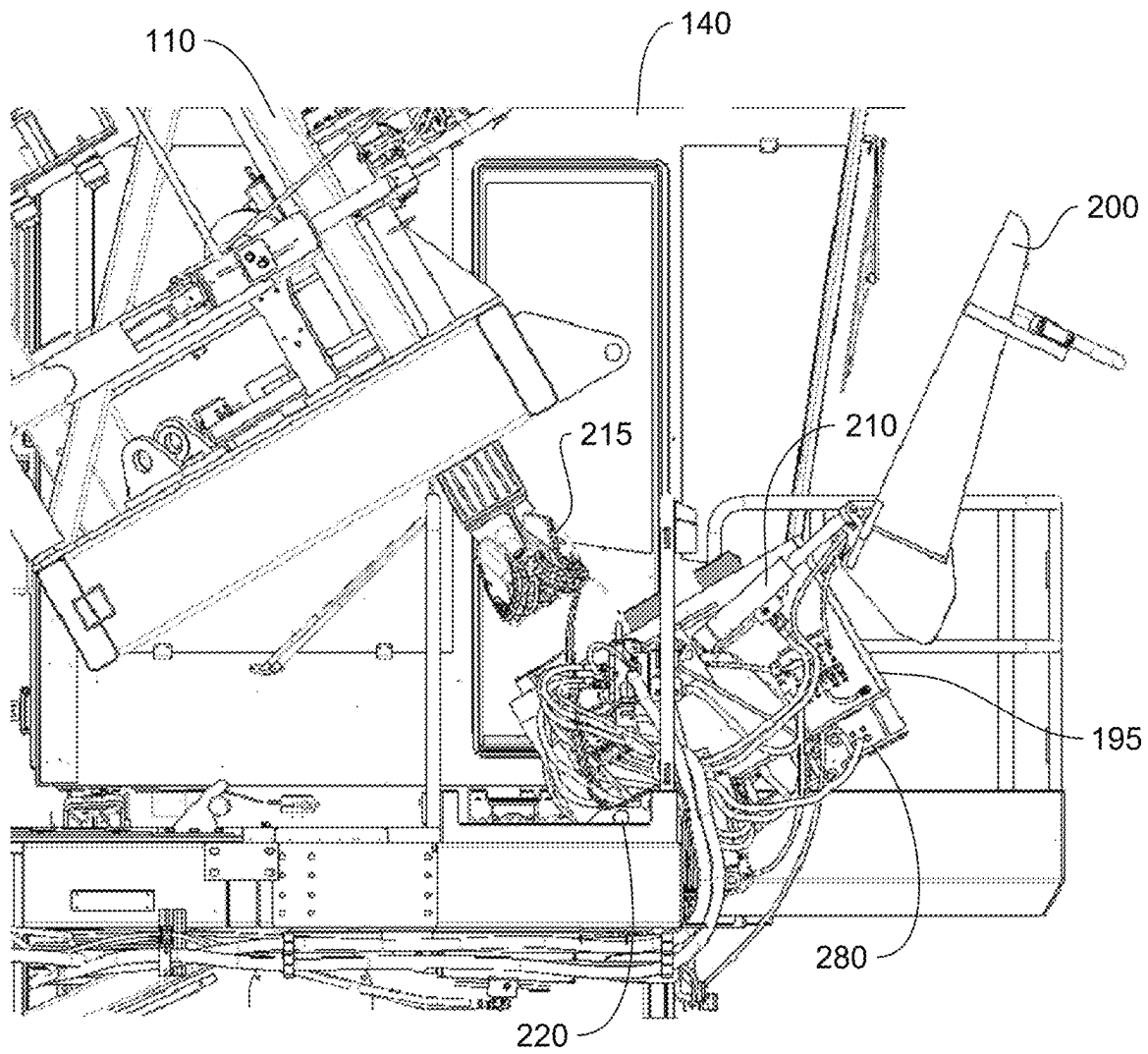


FIG. 5

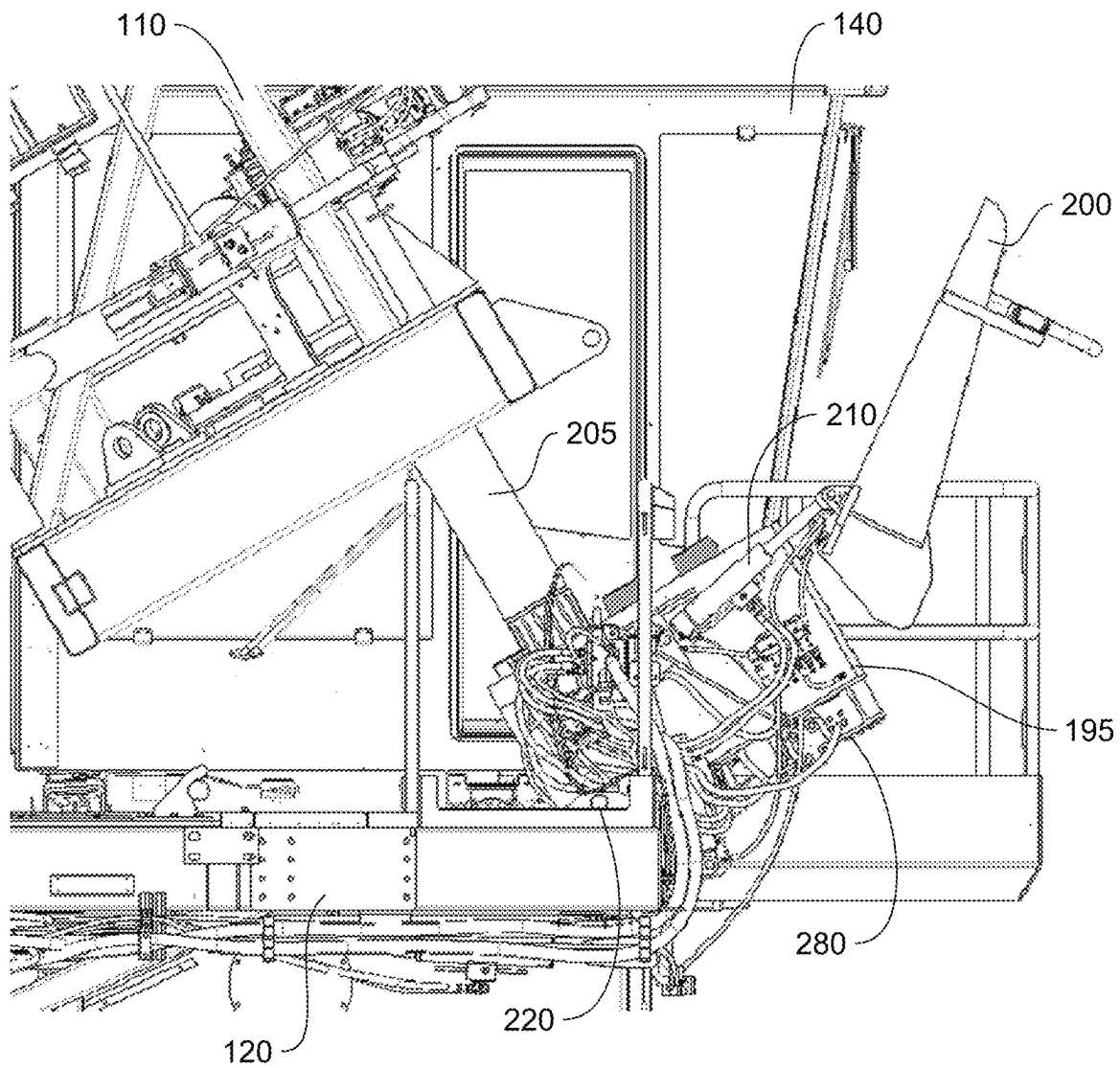


FIG. 6

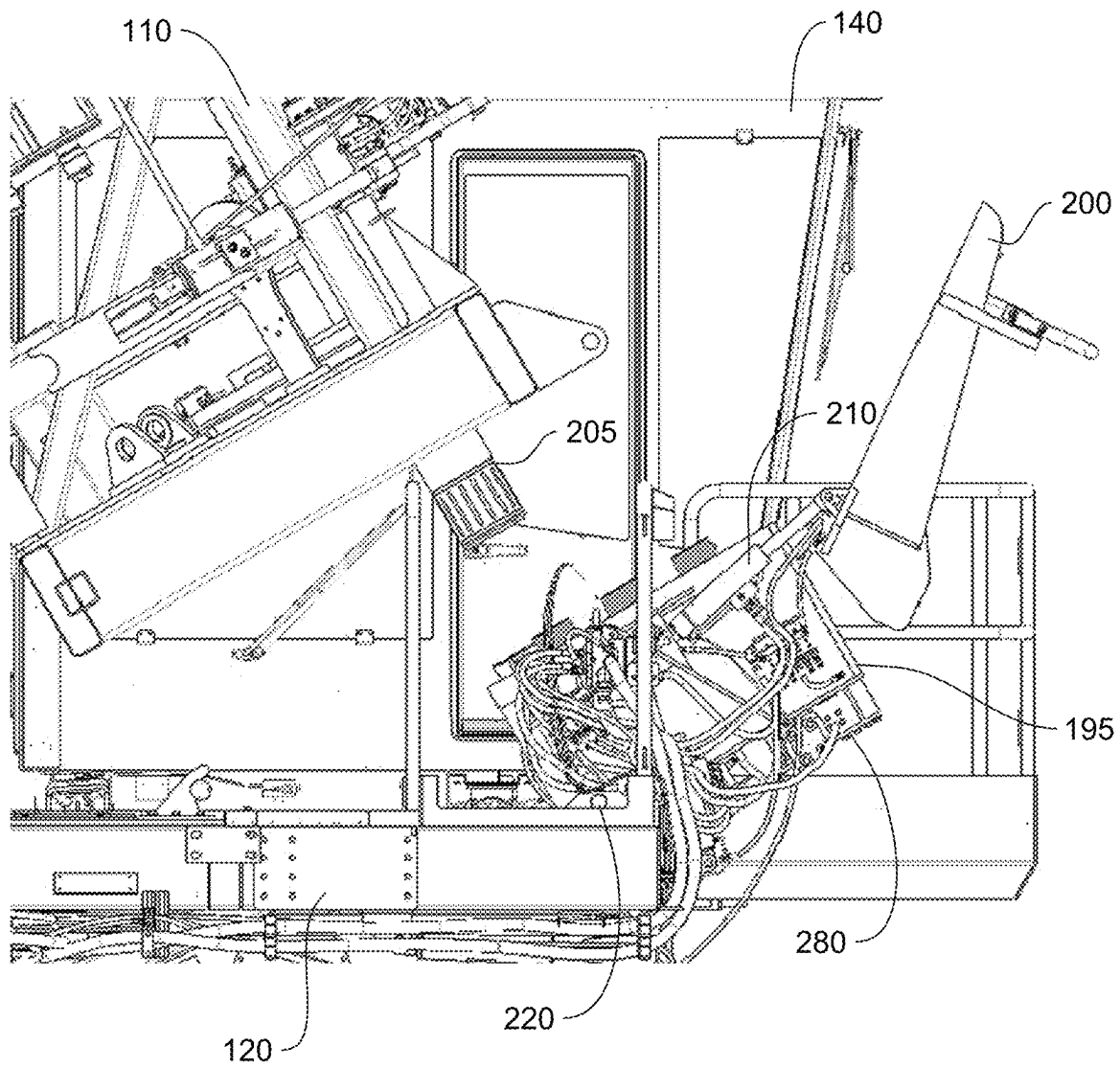
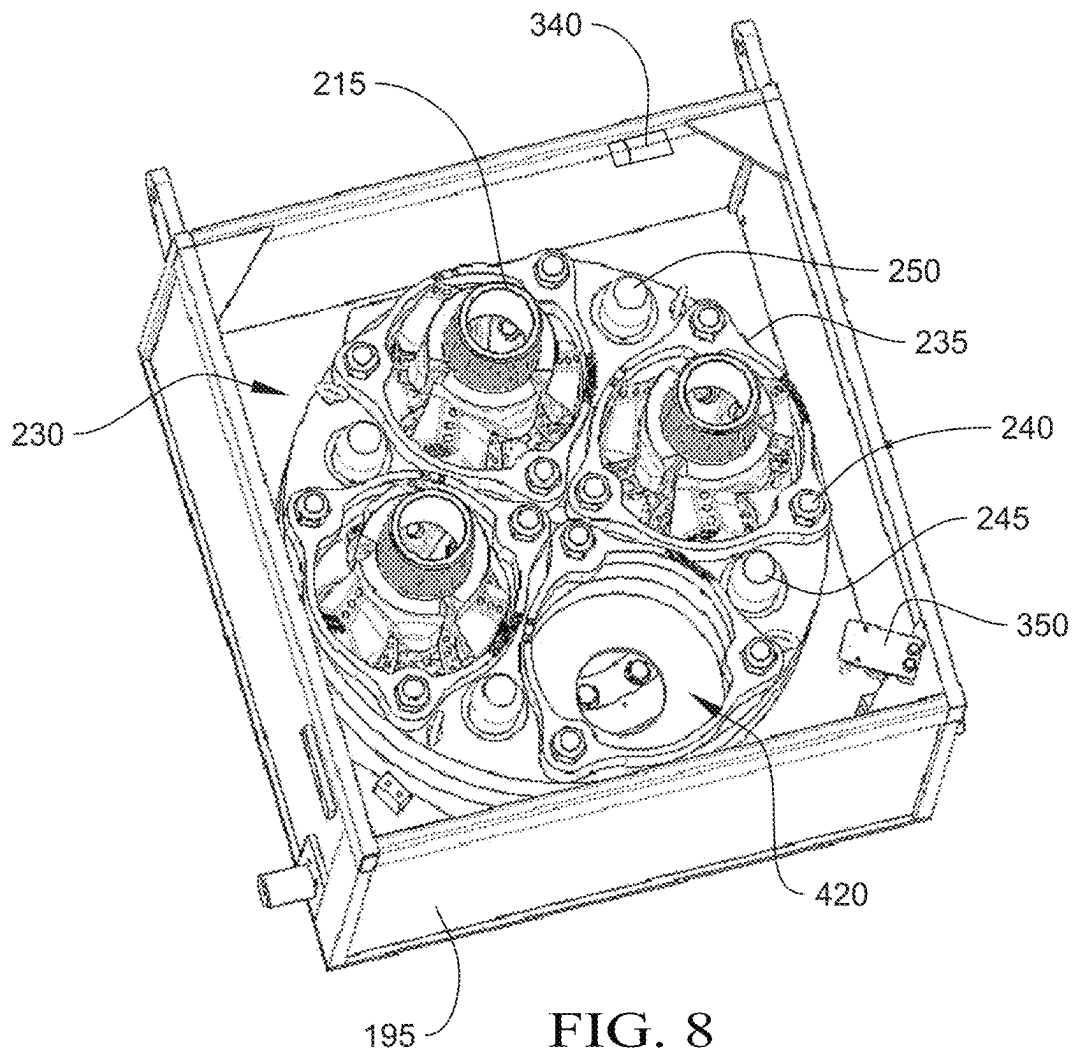


FIG. 7



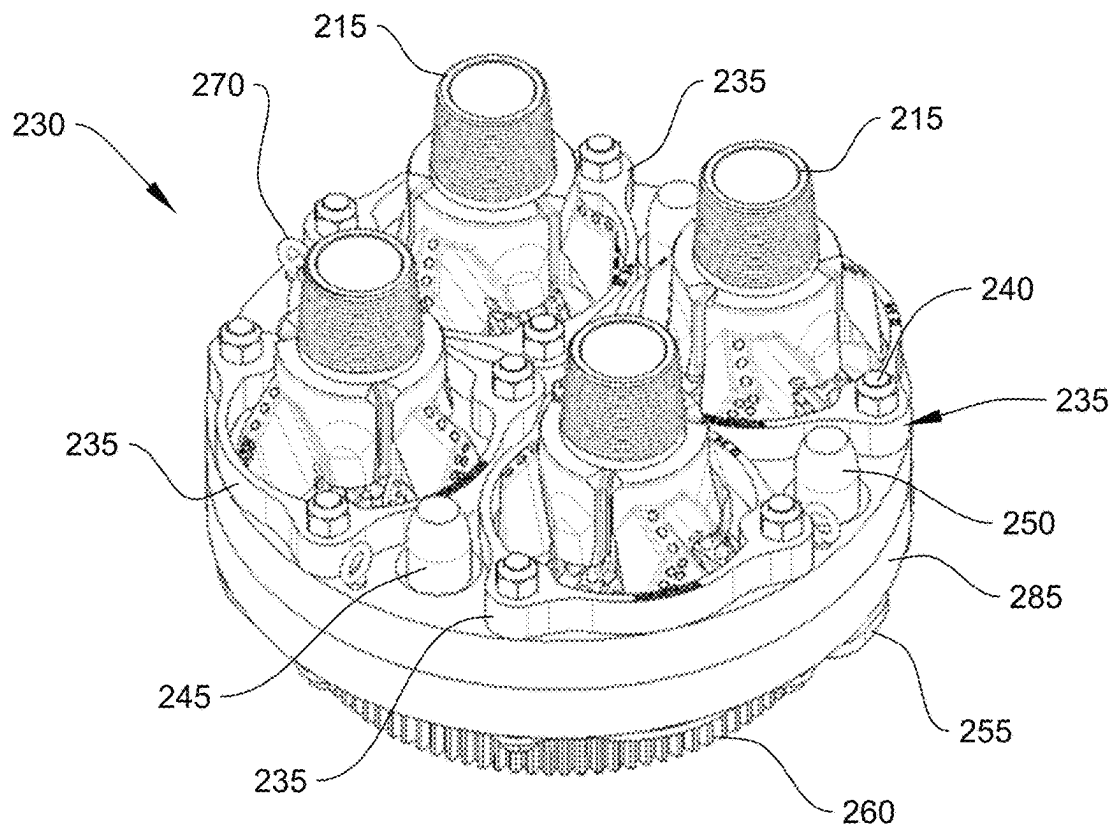


FIG. 9A

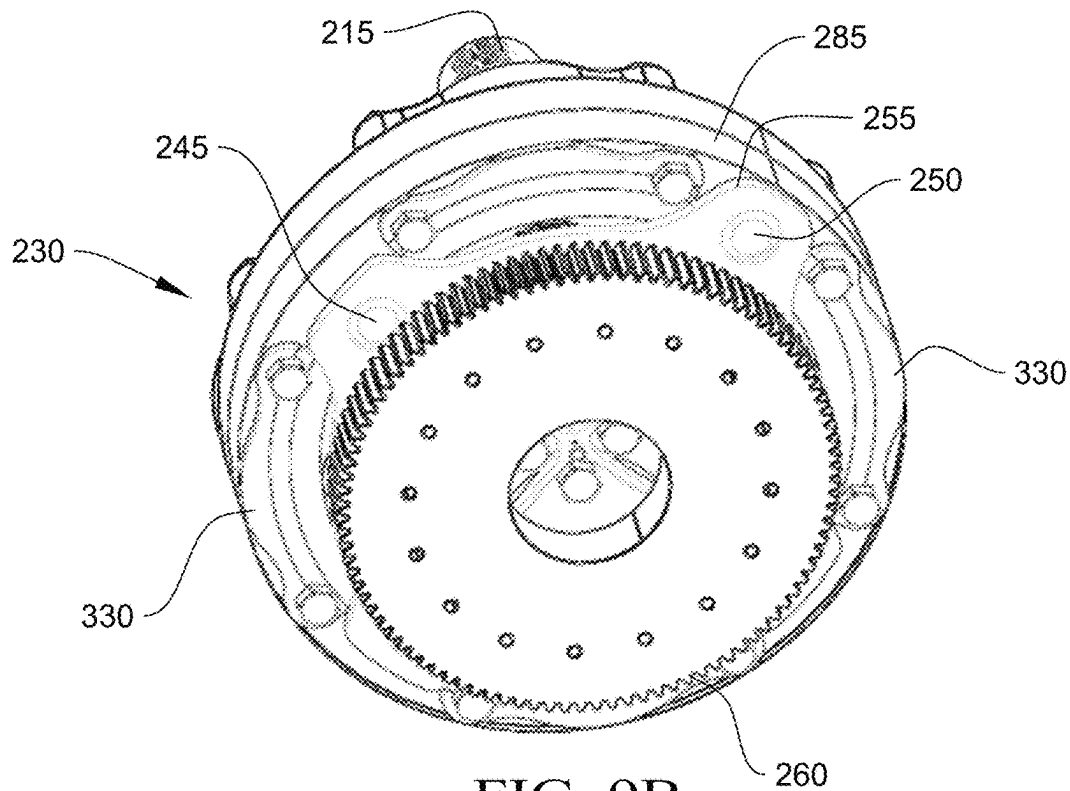


FIG. 9B

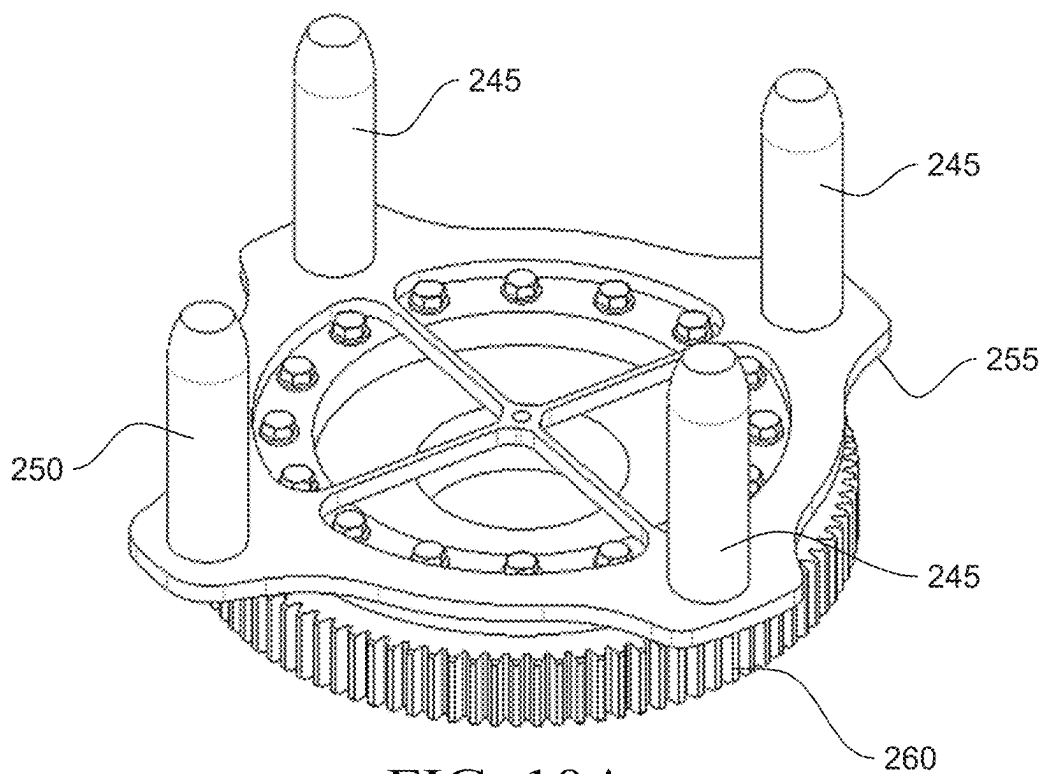


FIG. 10A

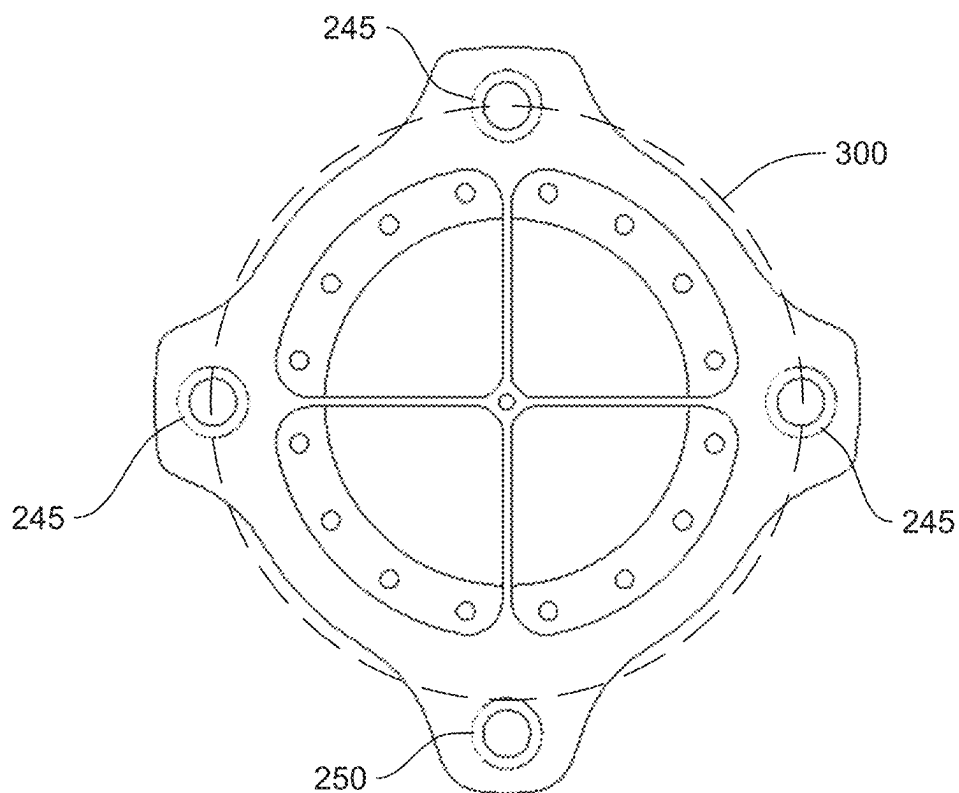


FIG. 10B

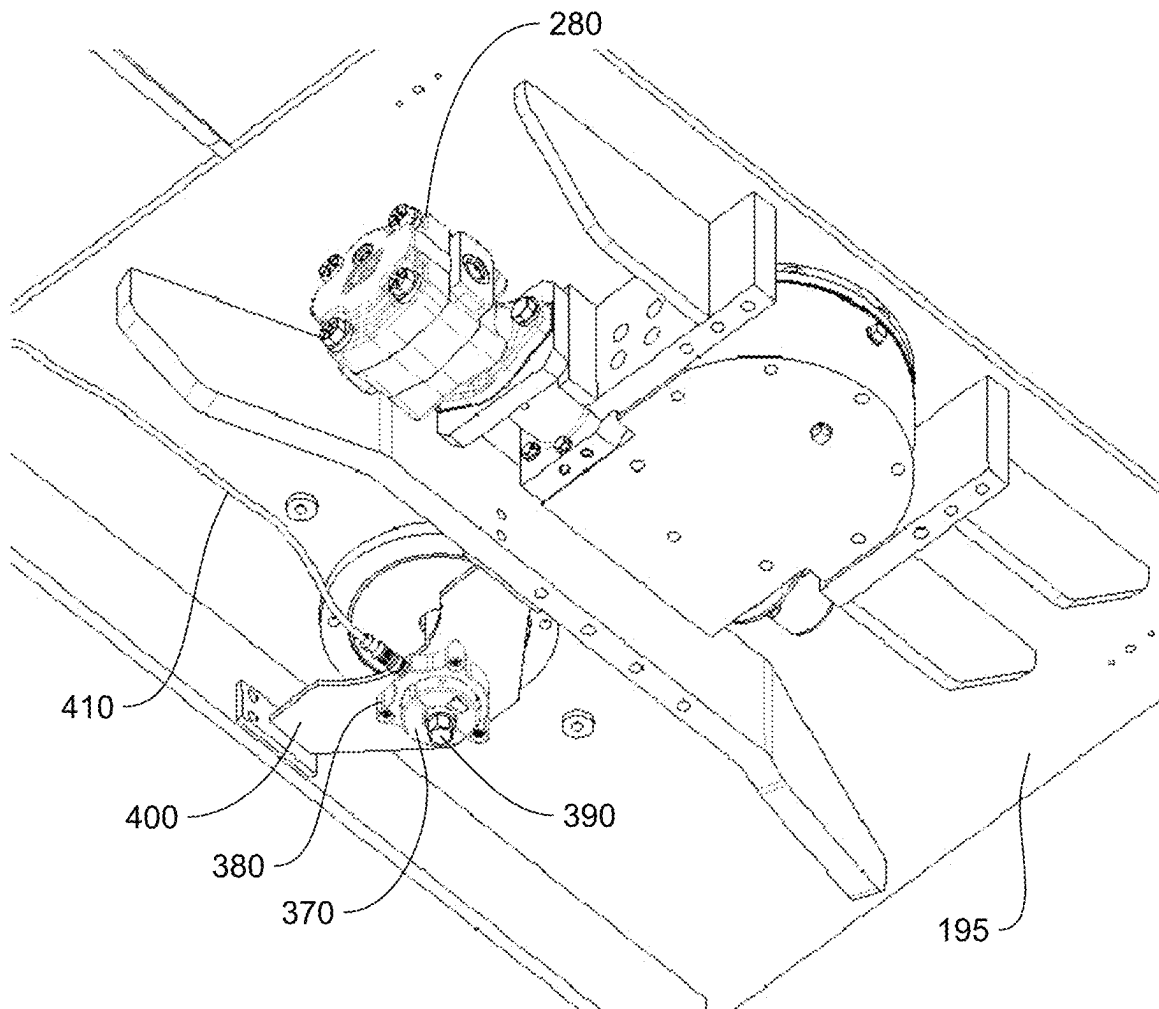


FIG. 11

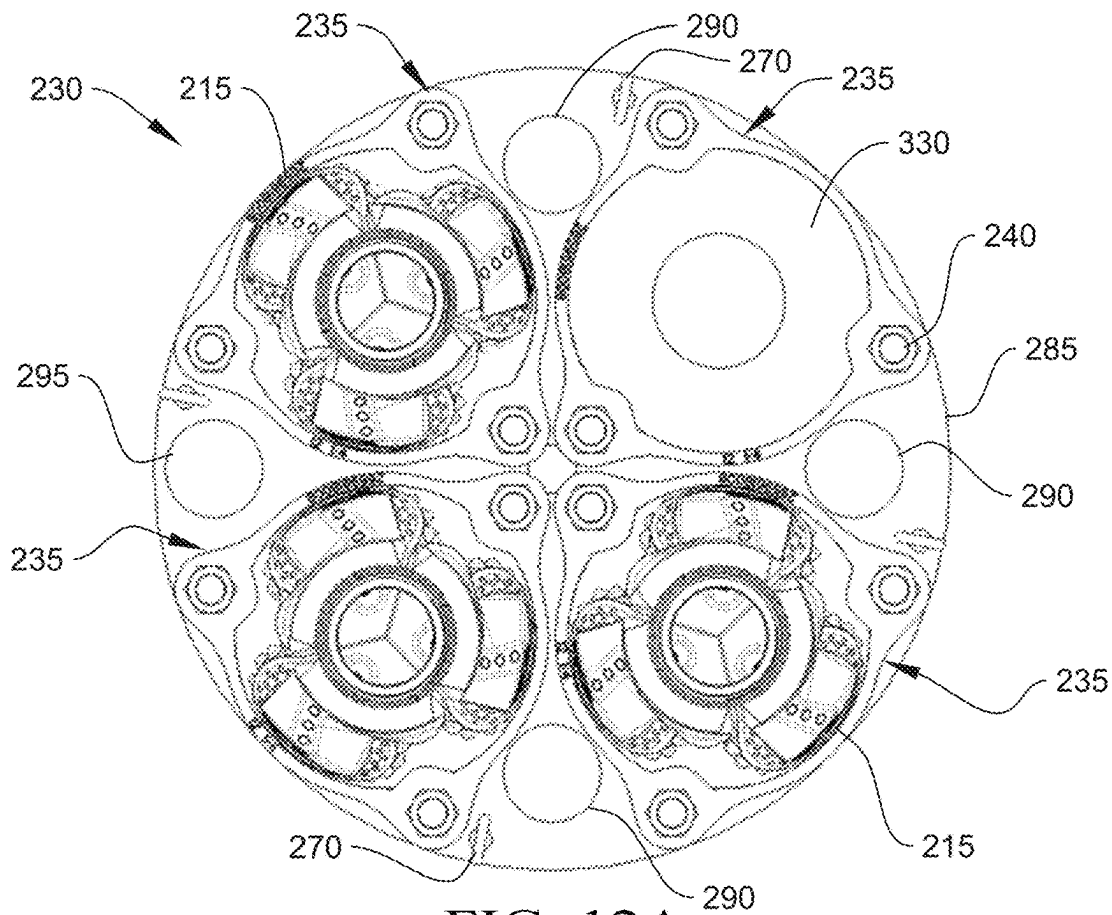


FIG. 12A

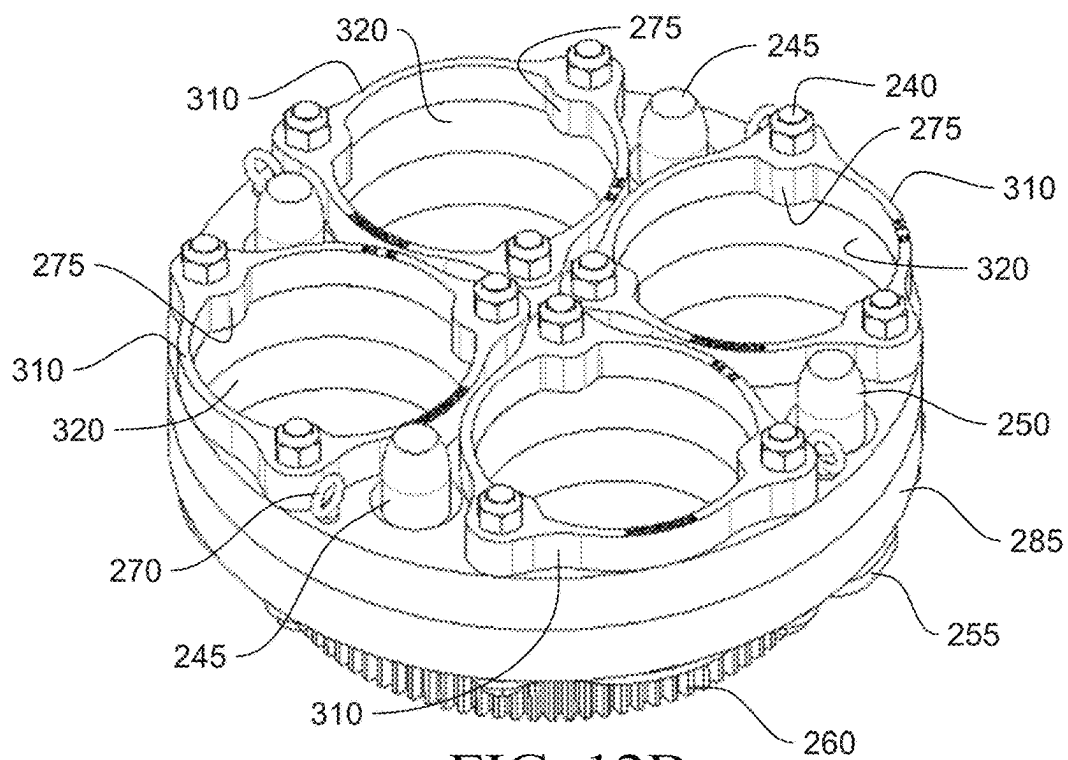


FIG. 12B

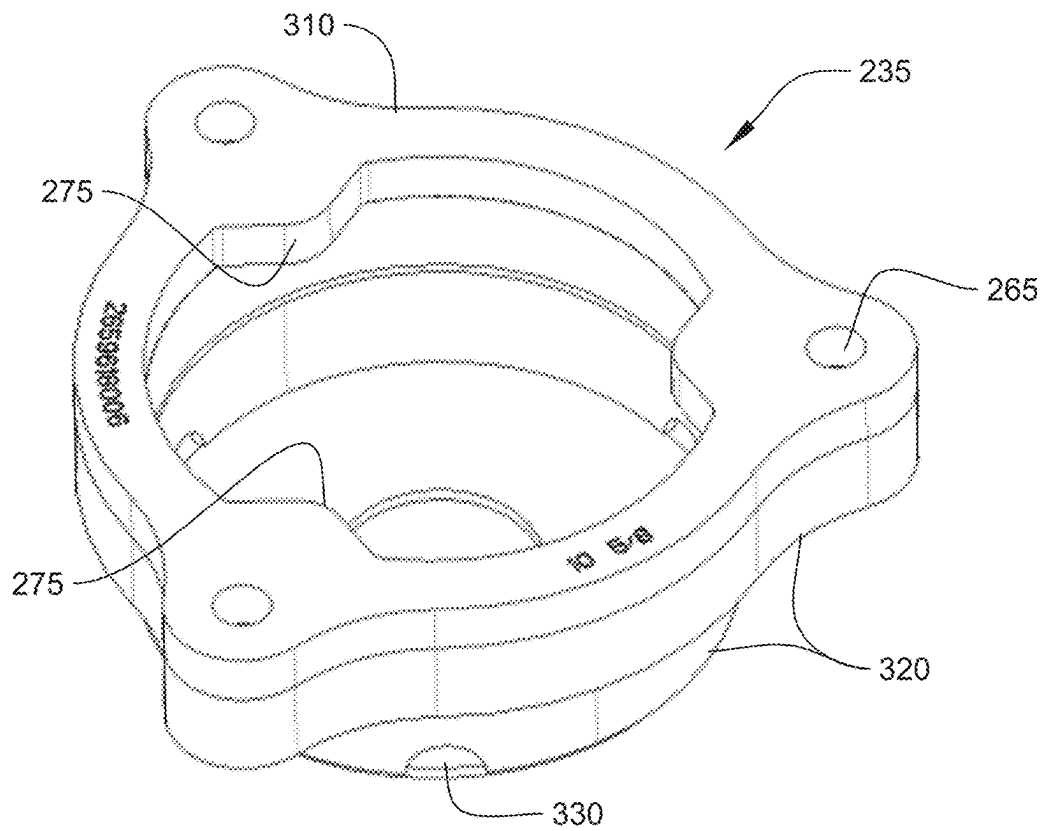


FIG. 13A

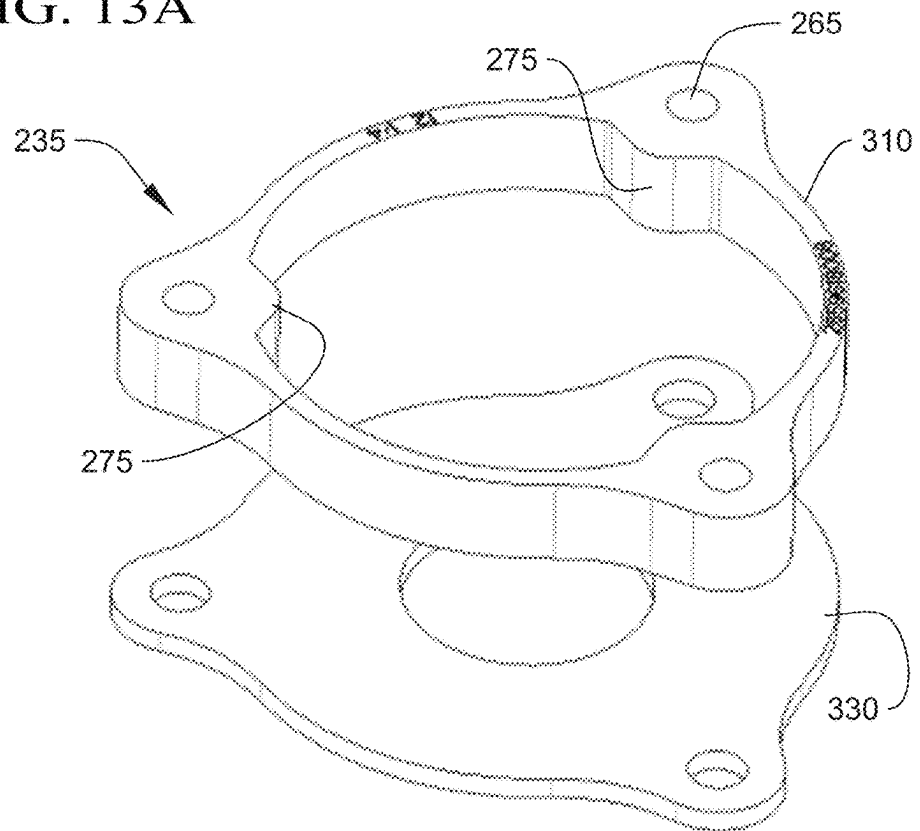


FIG. 13B

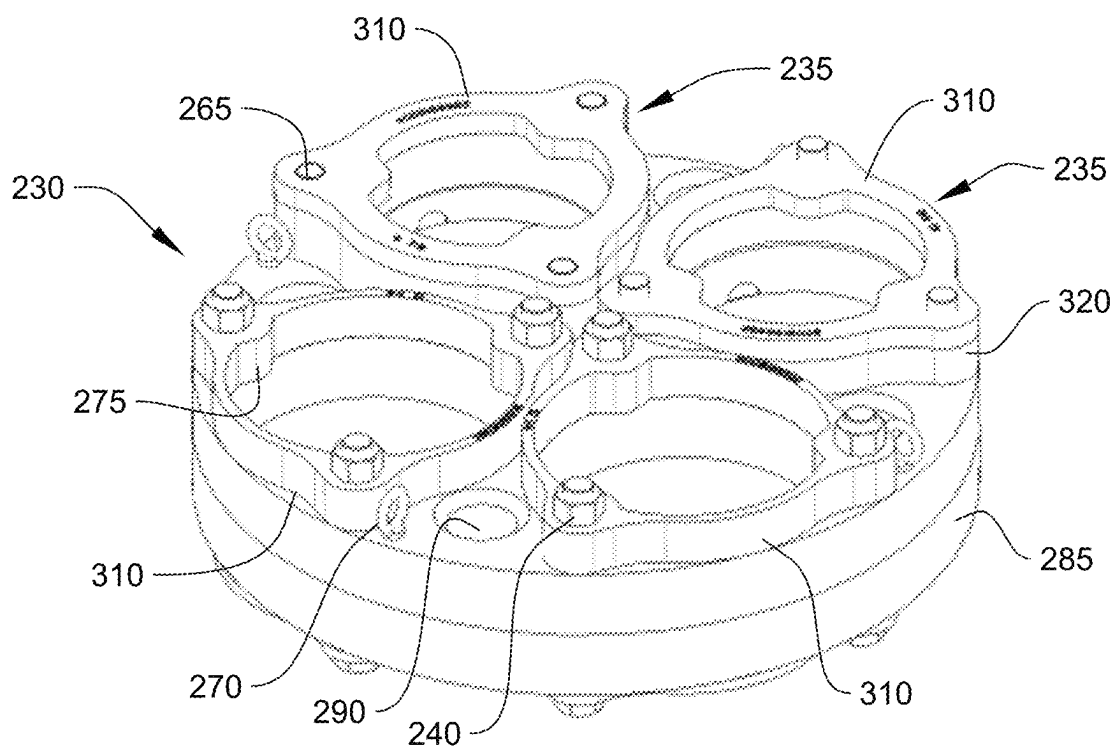


FIG. 14A

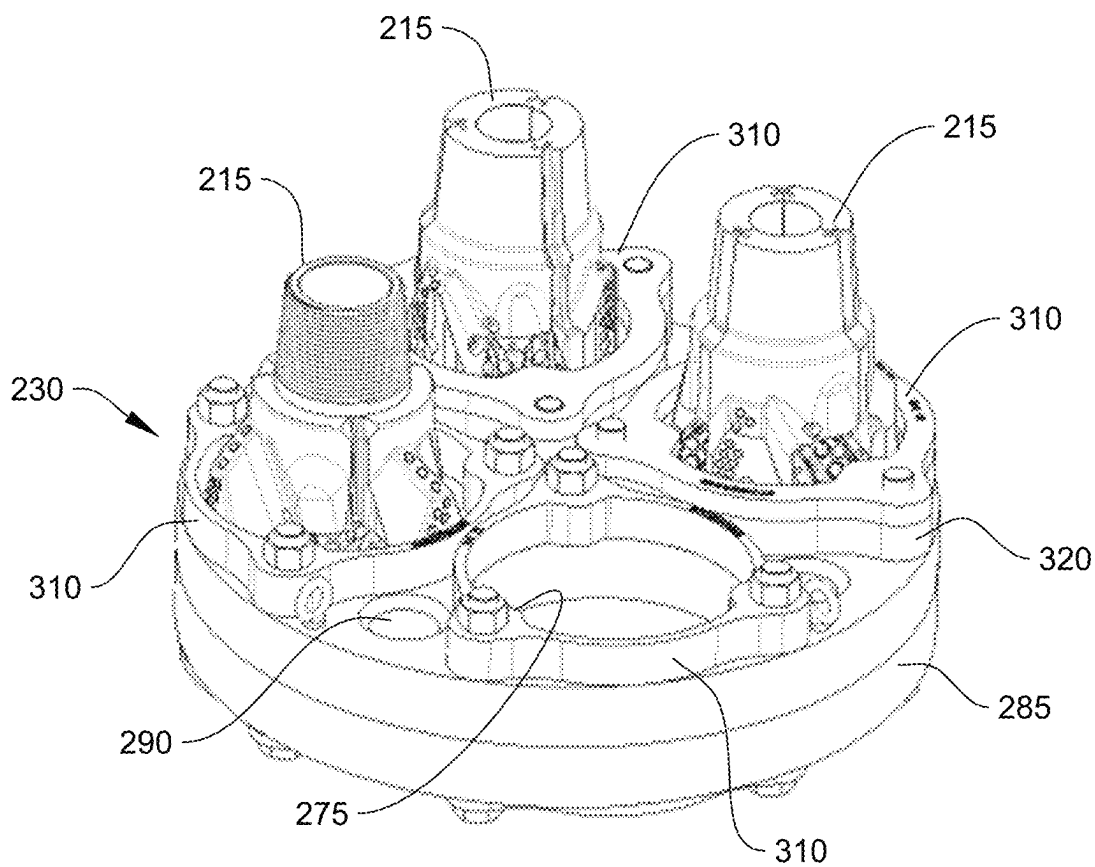


FIG. 14B

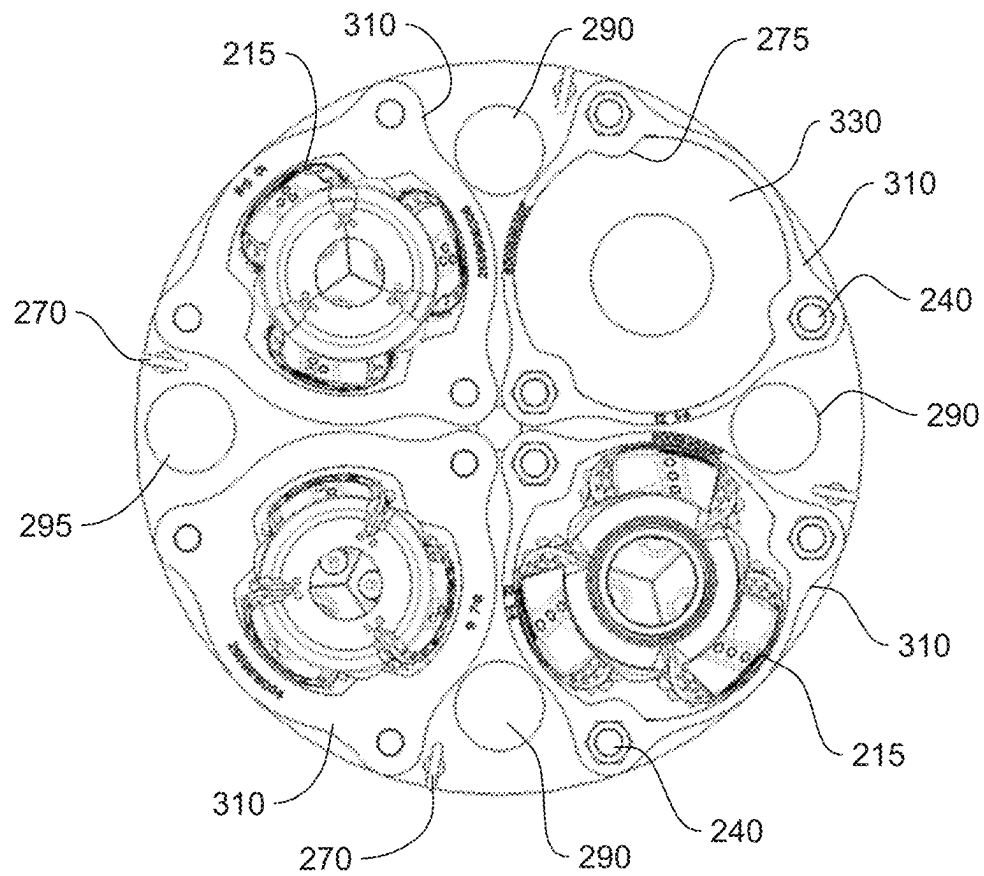


FIG. 15A

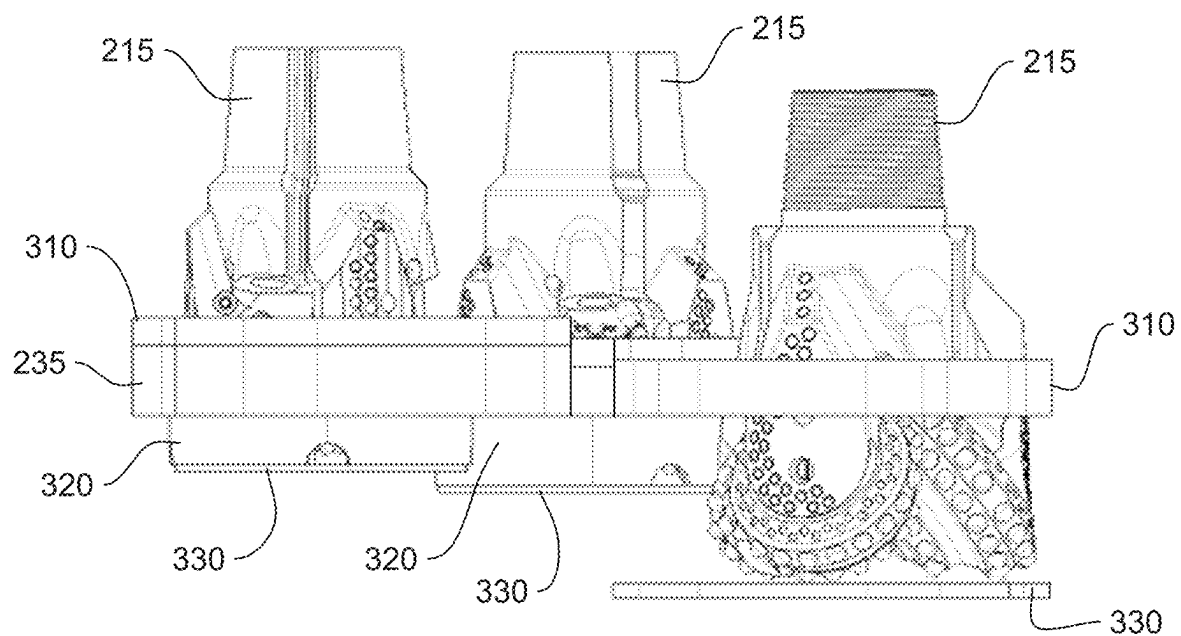


FIG. 15B

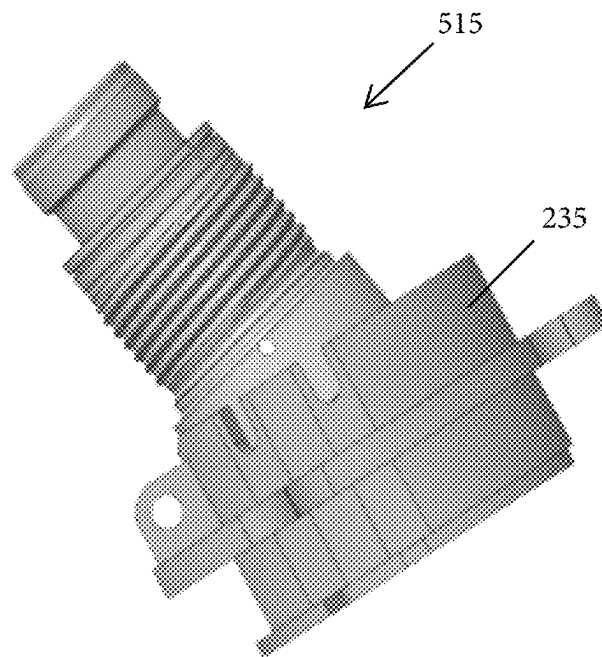


FIG. 16A

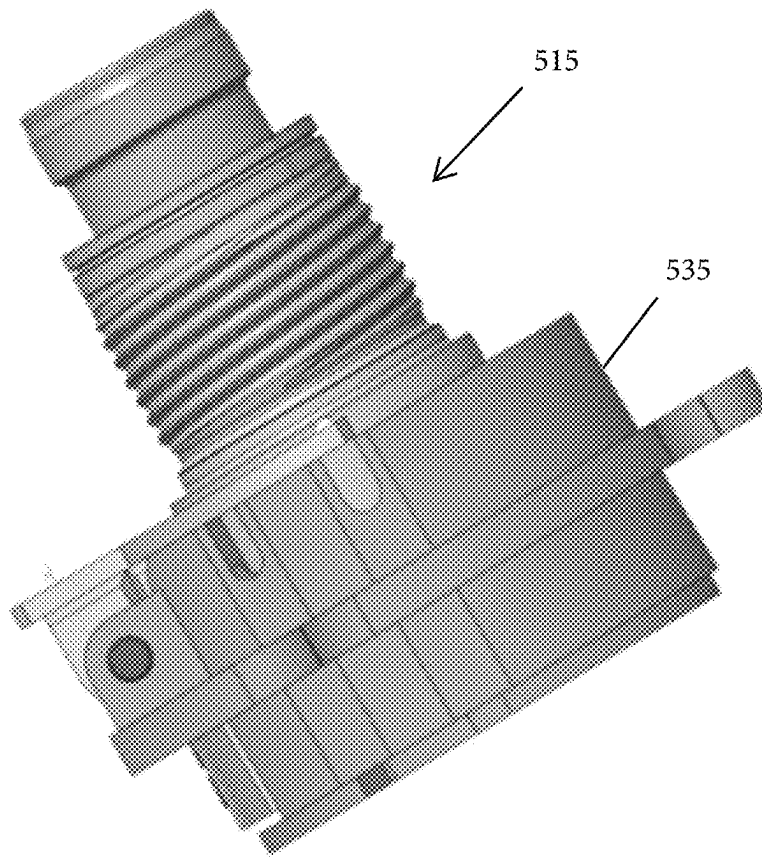


FIG. 16B

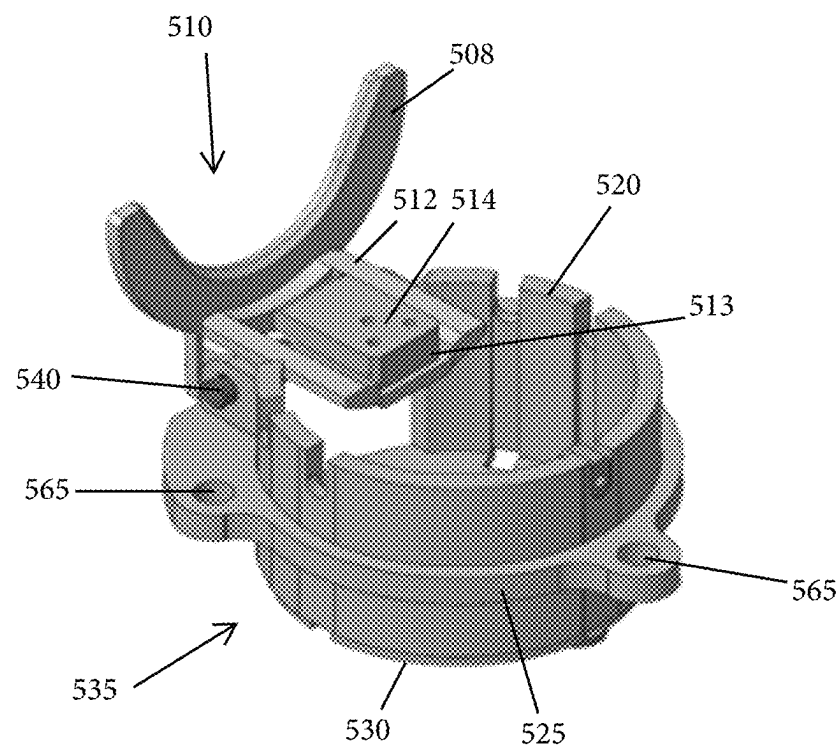


FIG. 17A

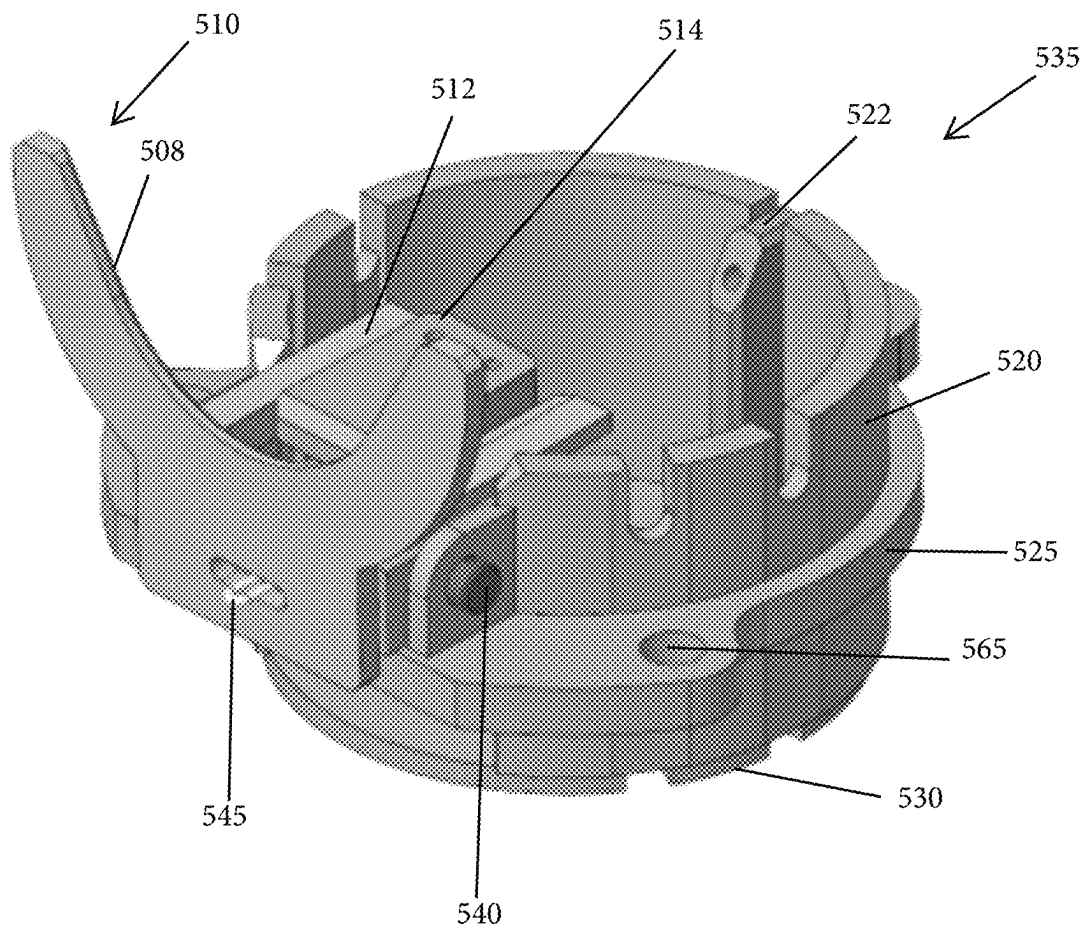


FIG. 17B

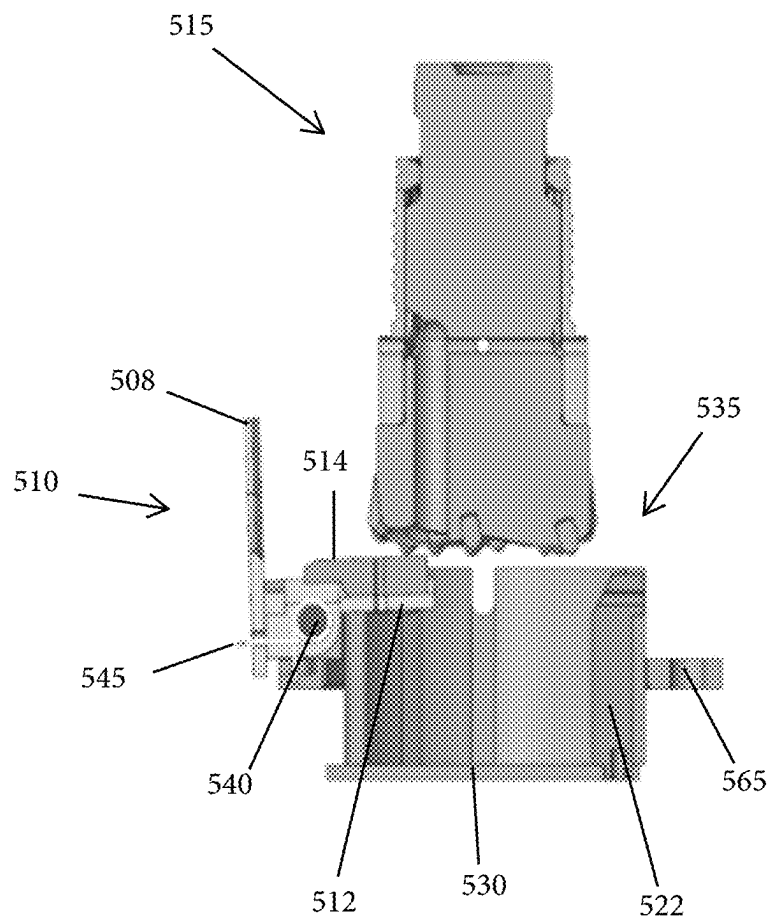


FIG. 18A

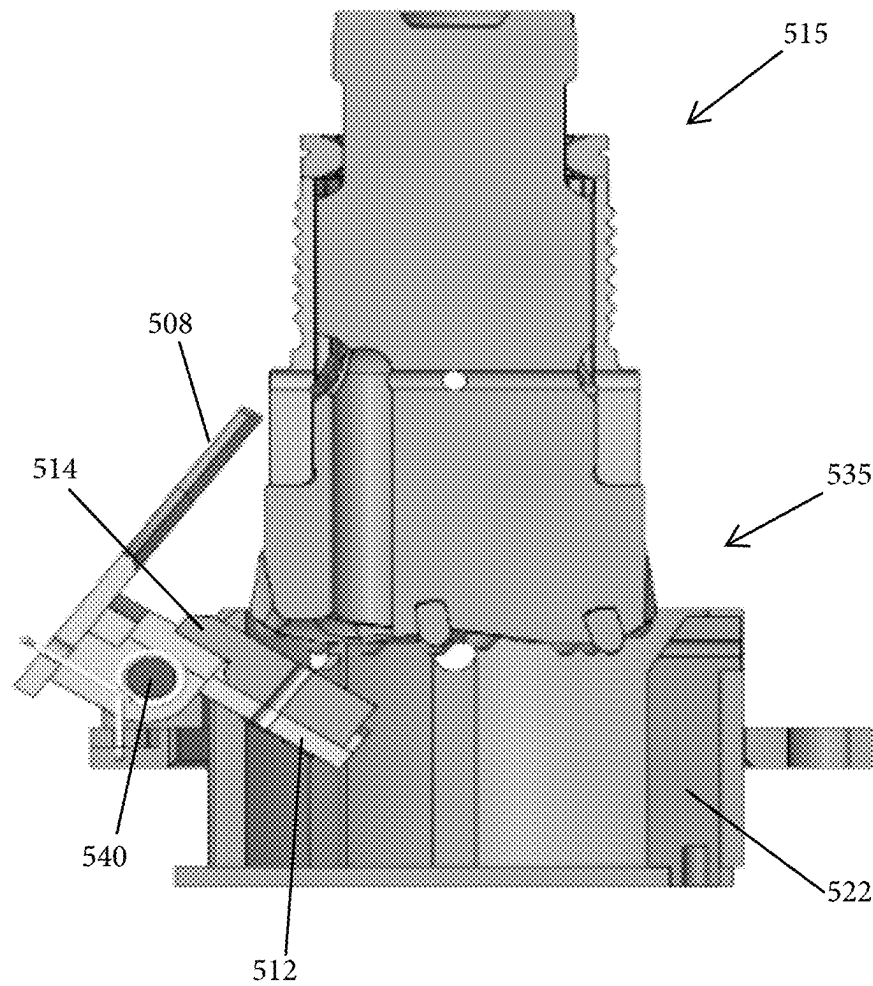


FIG. 18B

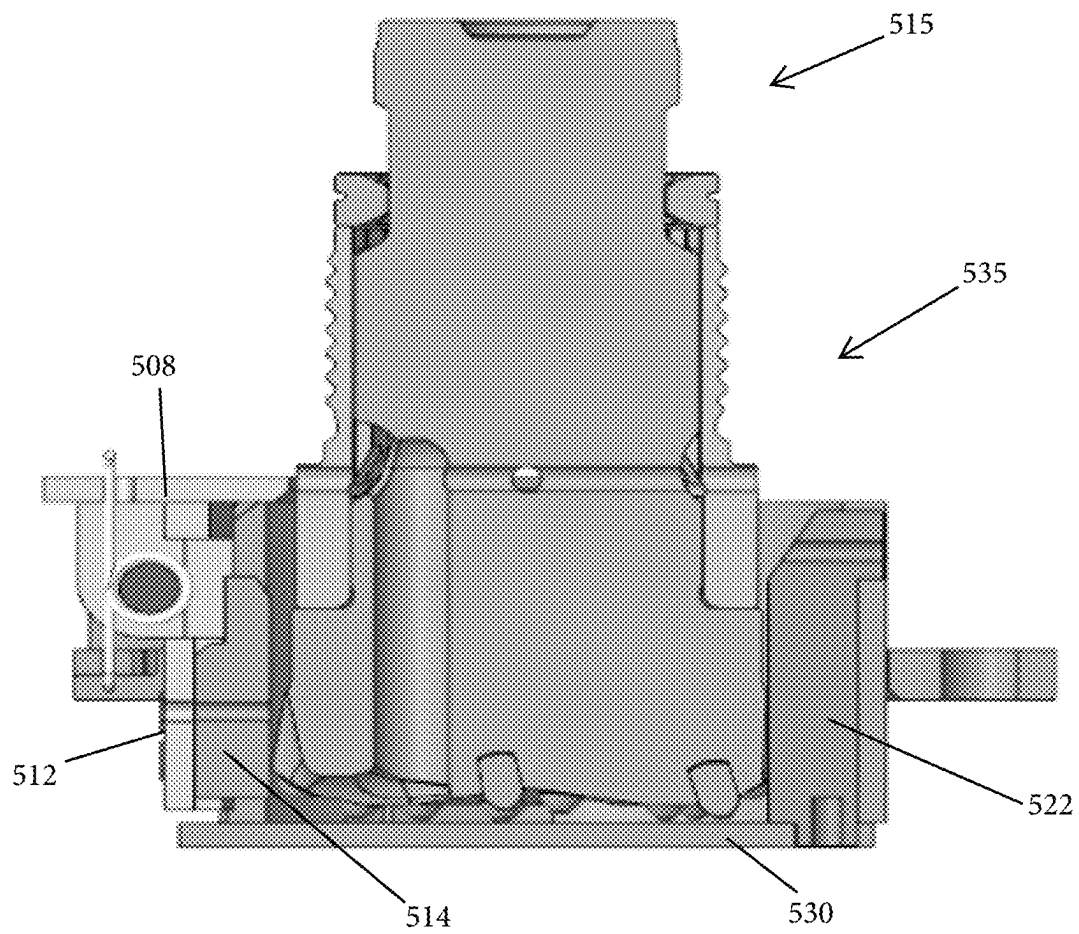


FIG. 18C

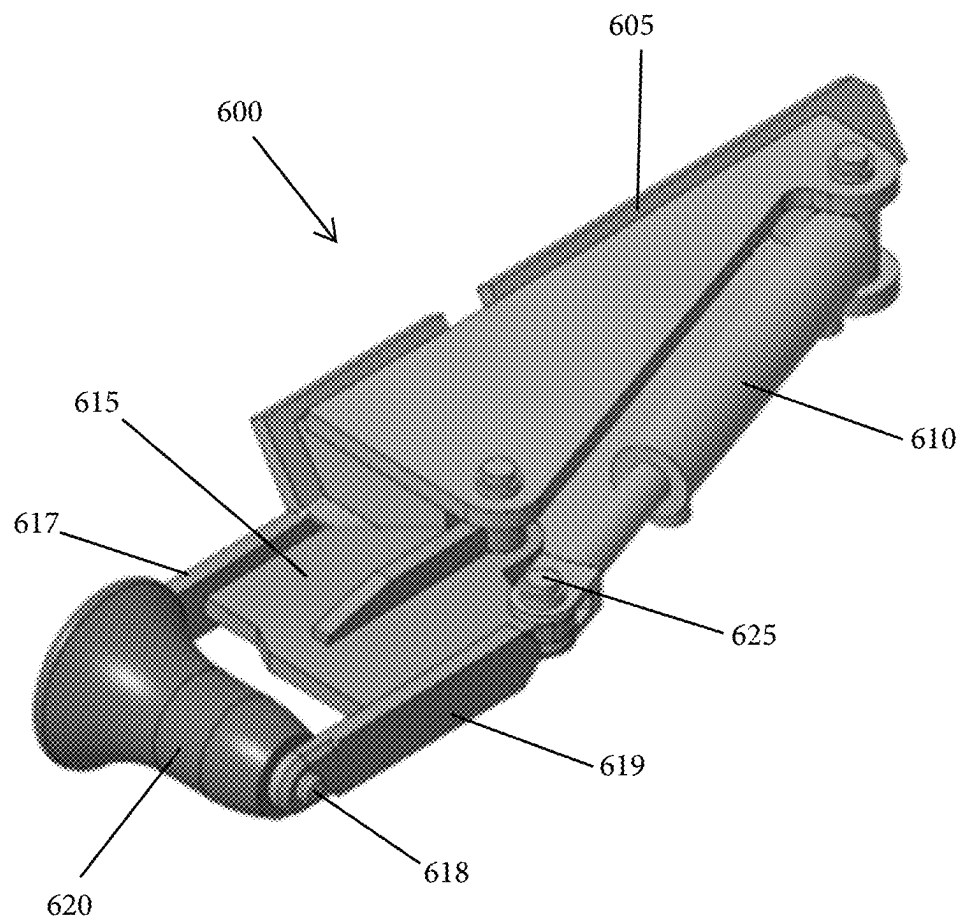


FIG. 19A

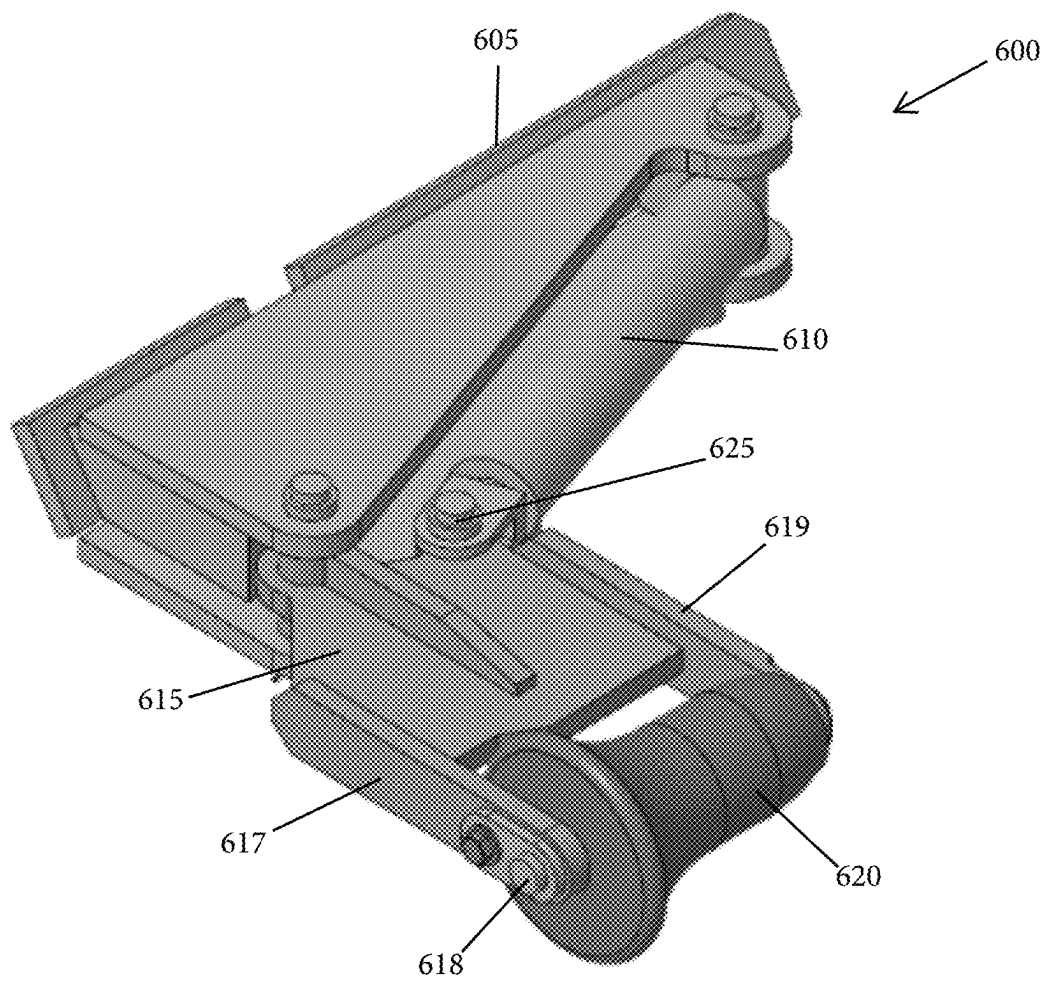


FIG. 19B

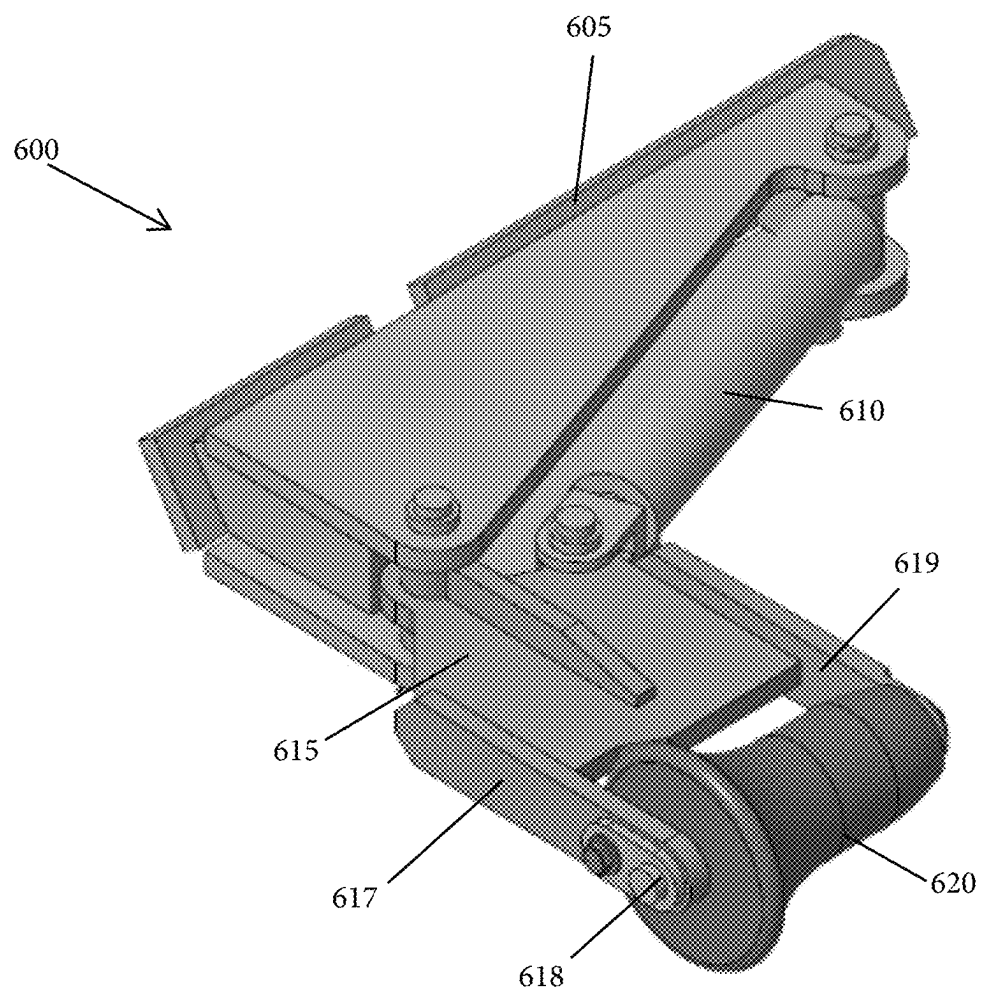


FIG. 20A

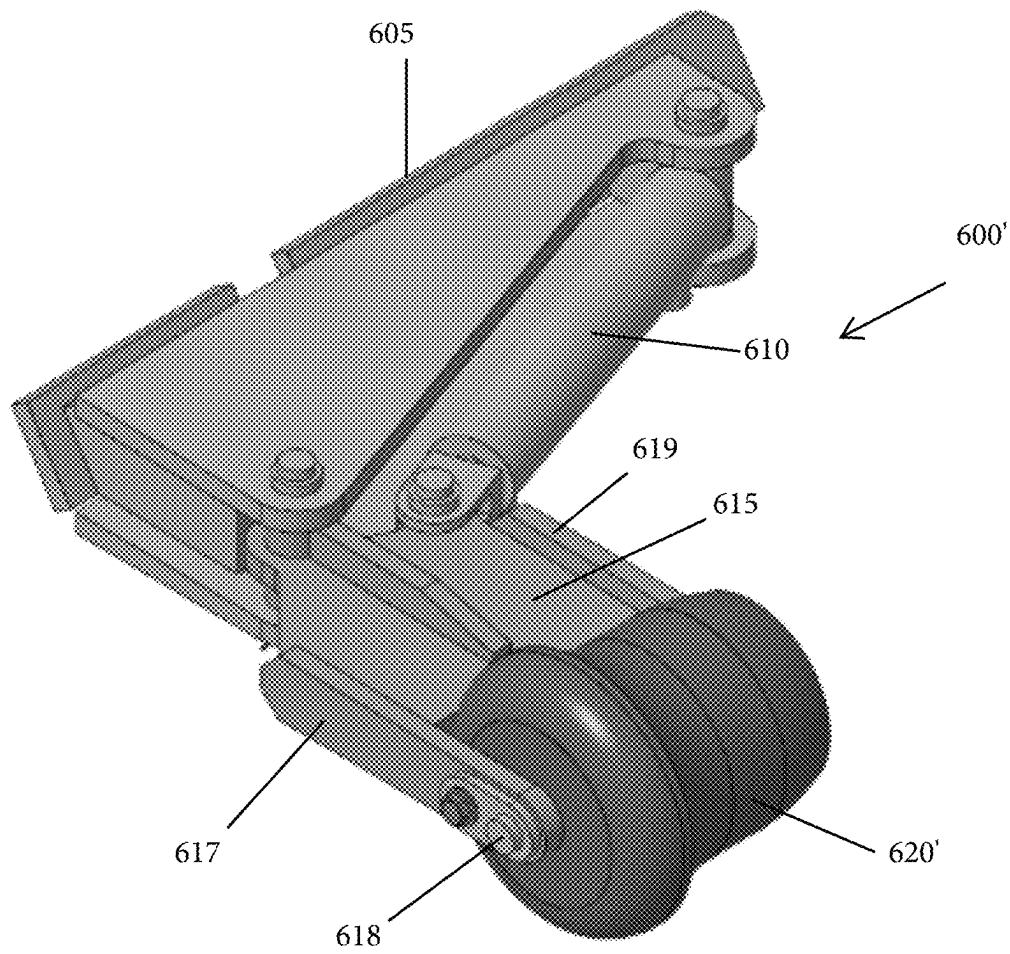


FIG. 20B

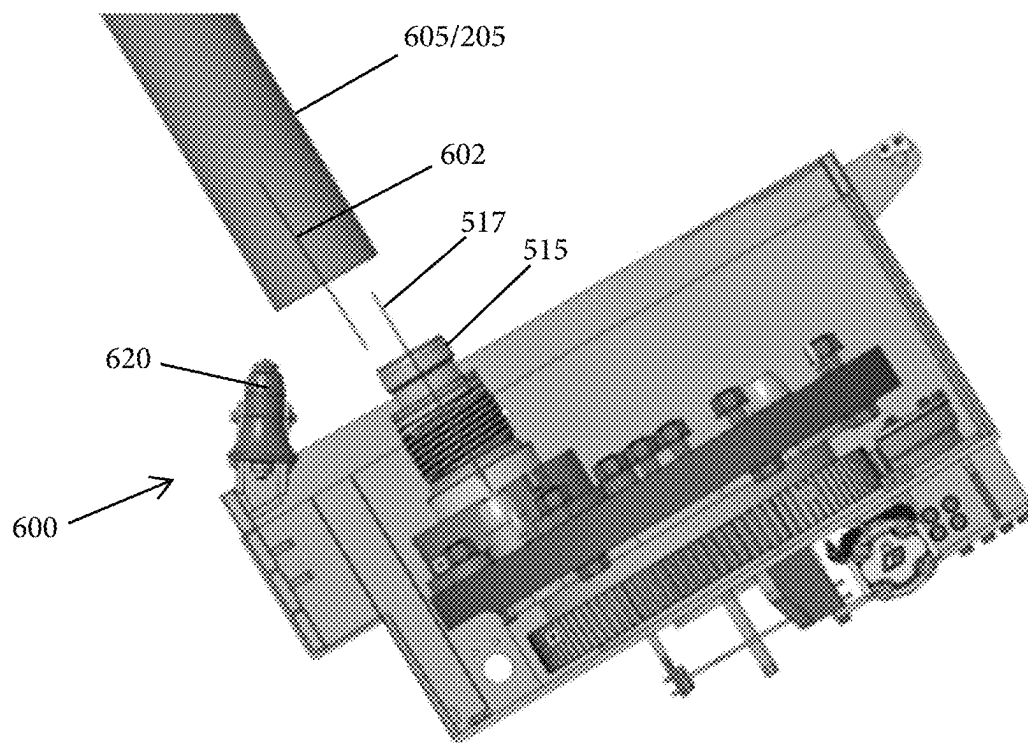


FIG. 21A

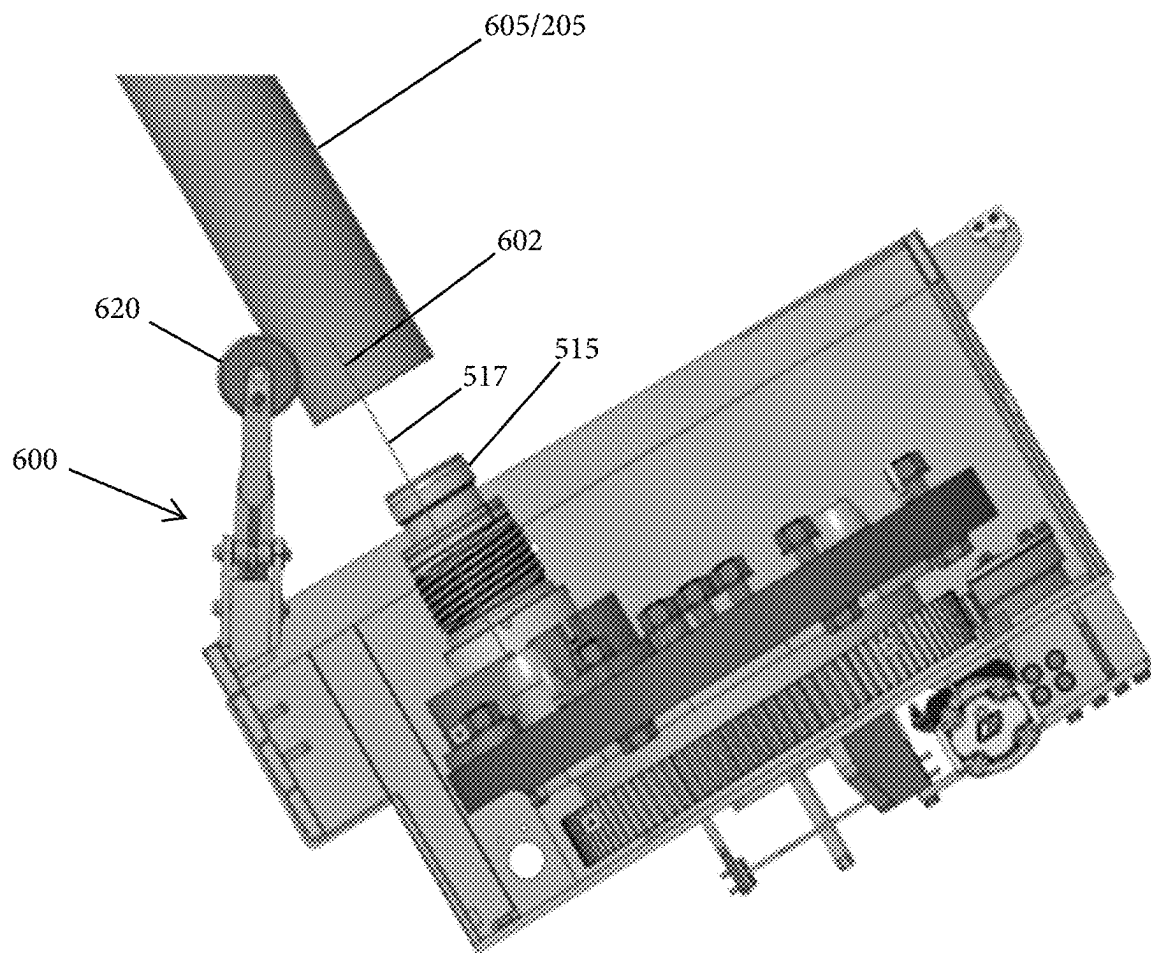
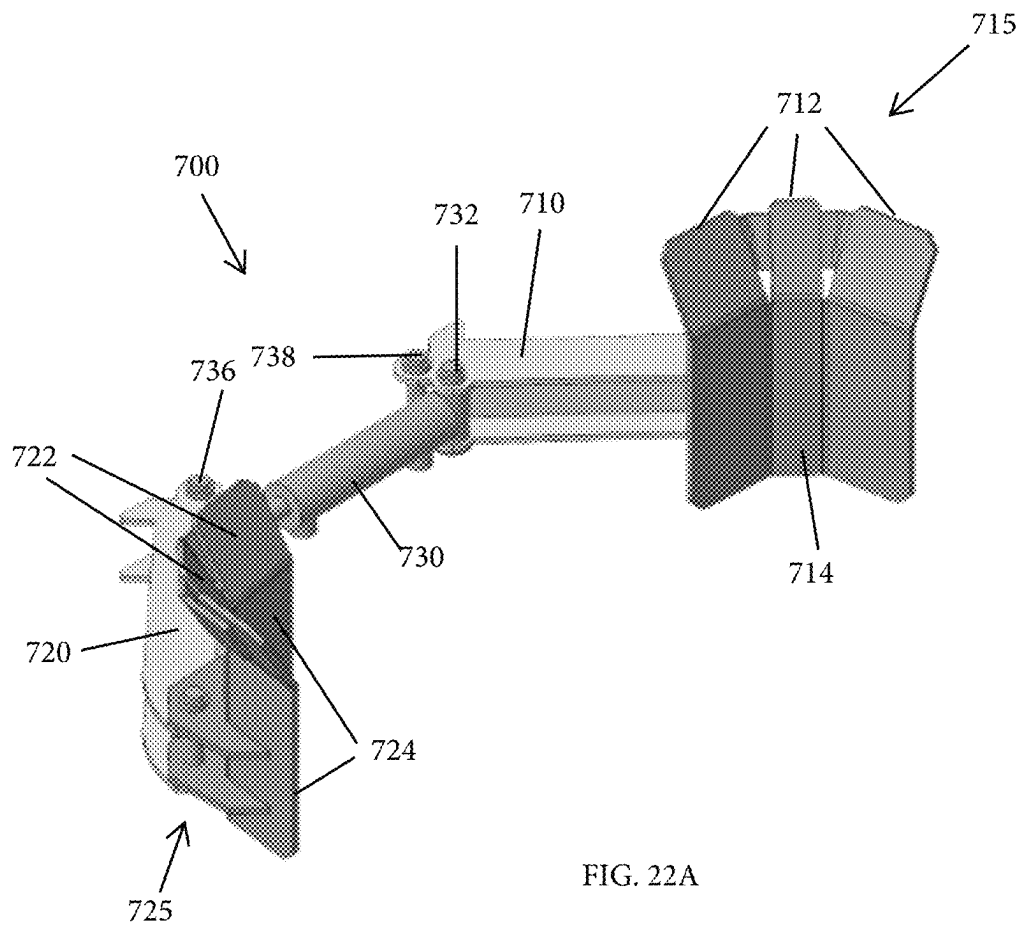


FIG. 21B



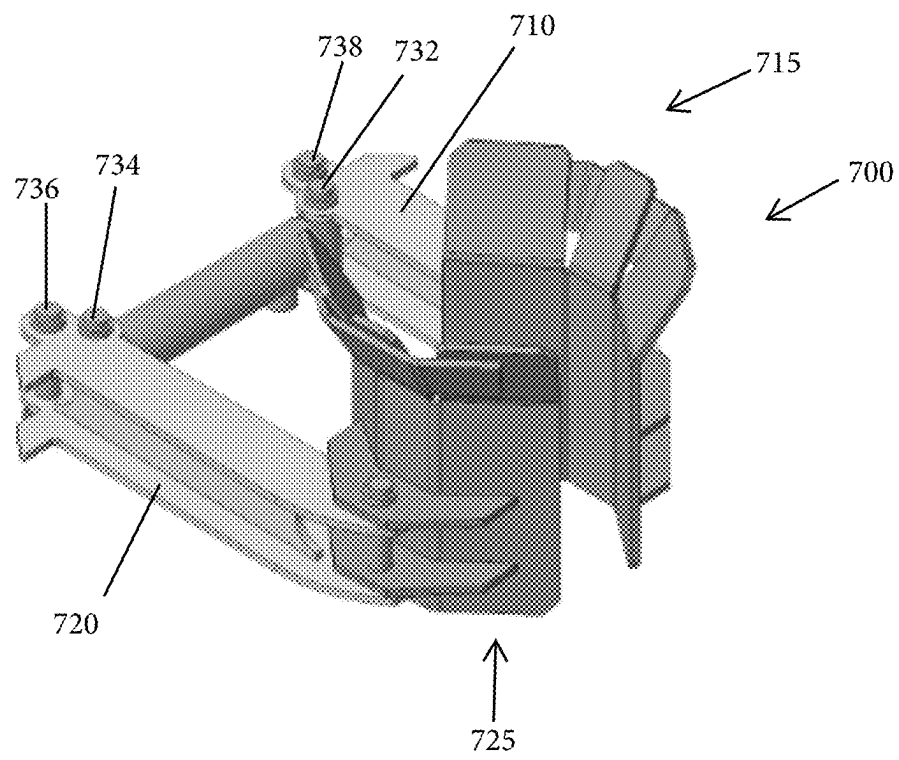


FIG. 22B

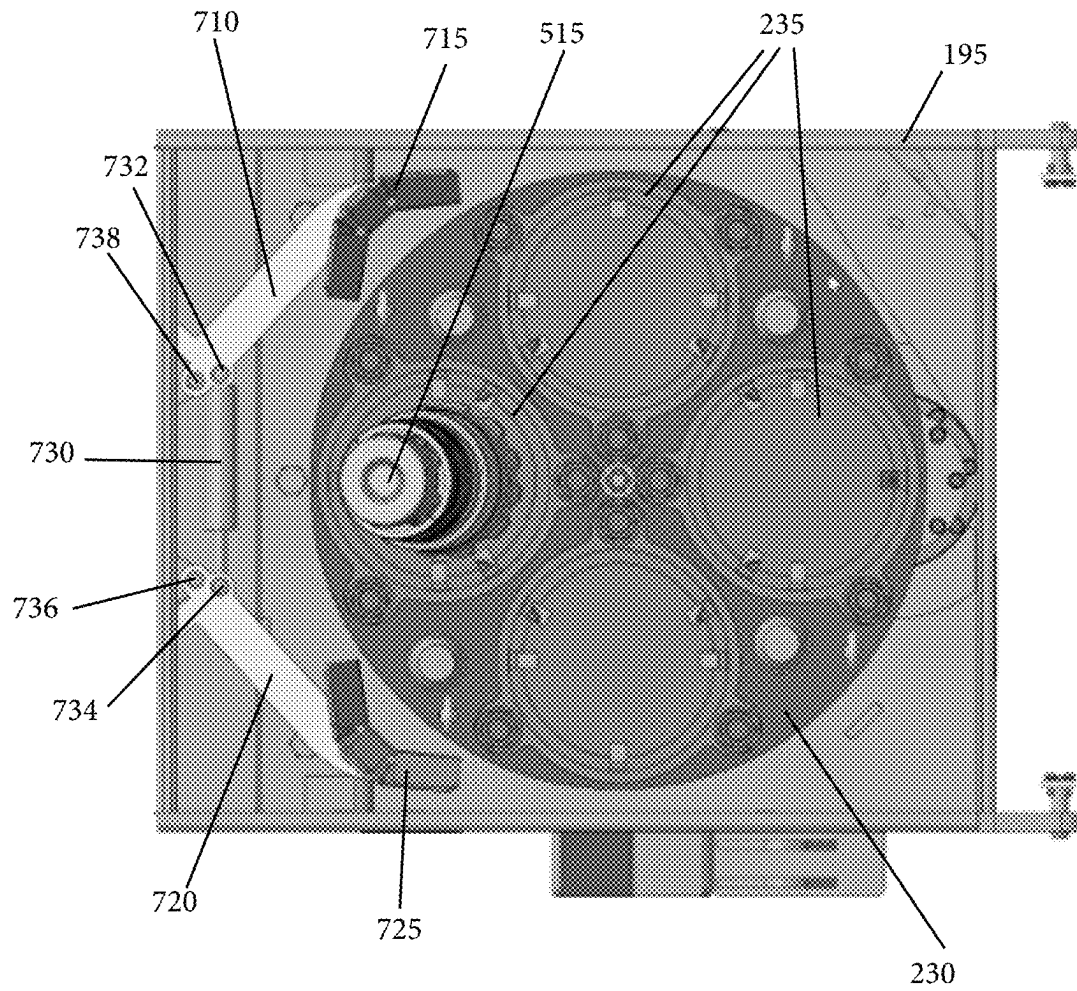


FIG. 23A

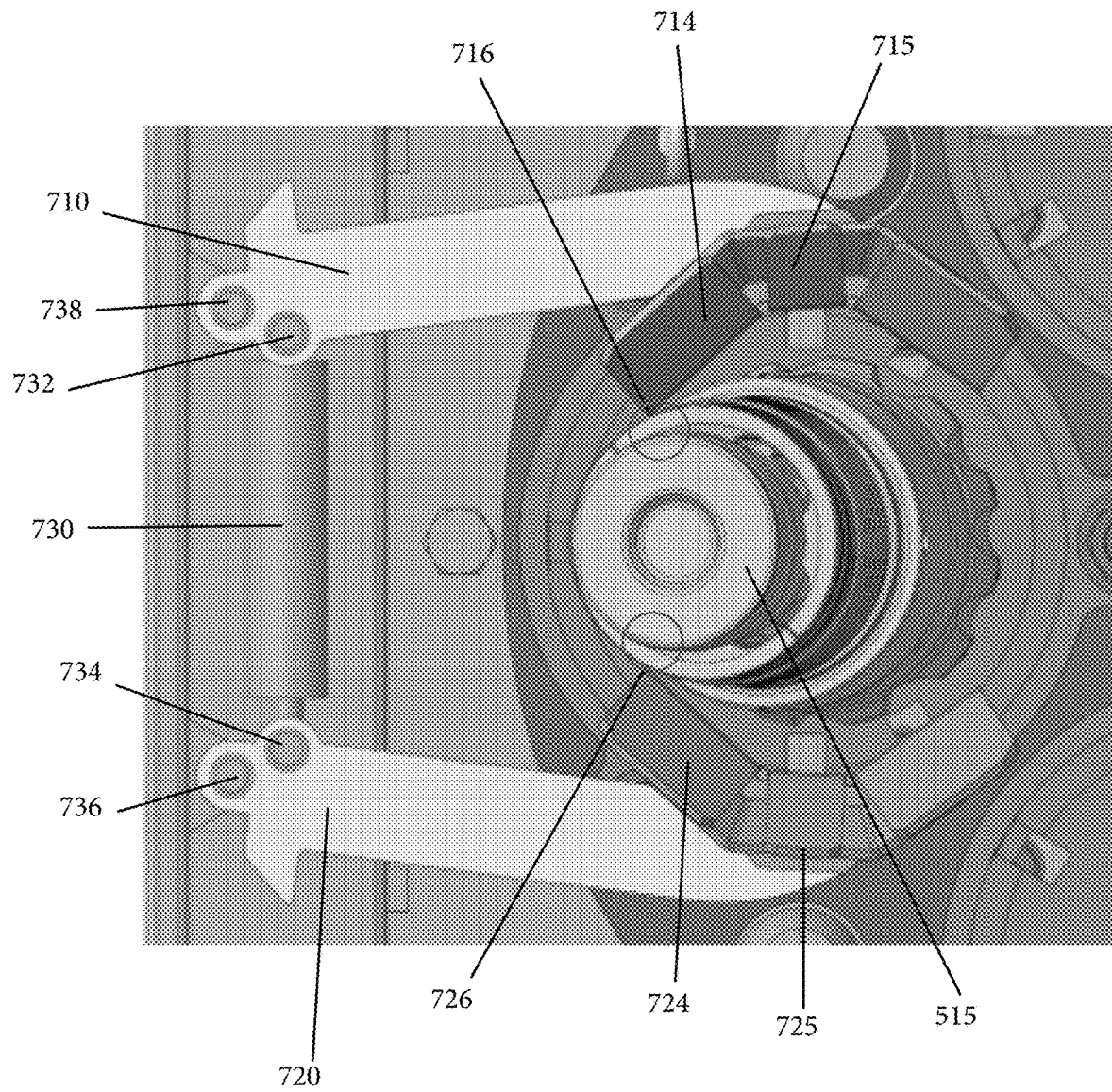


FIG. 23B

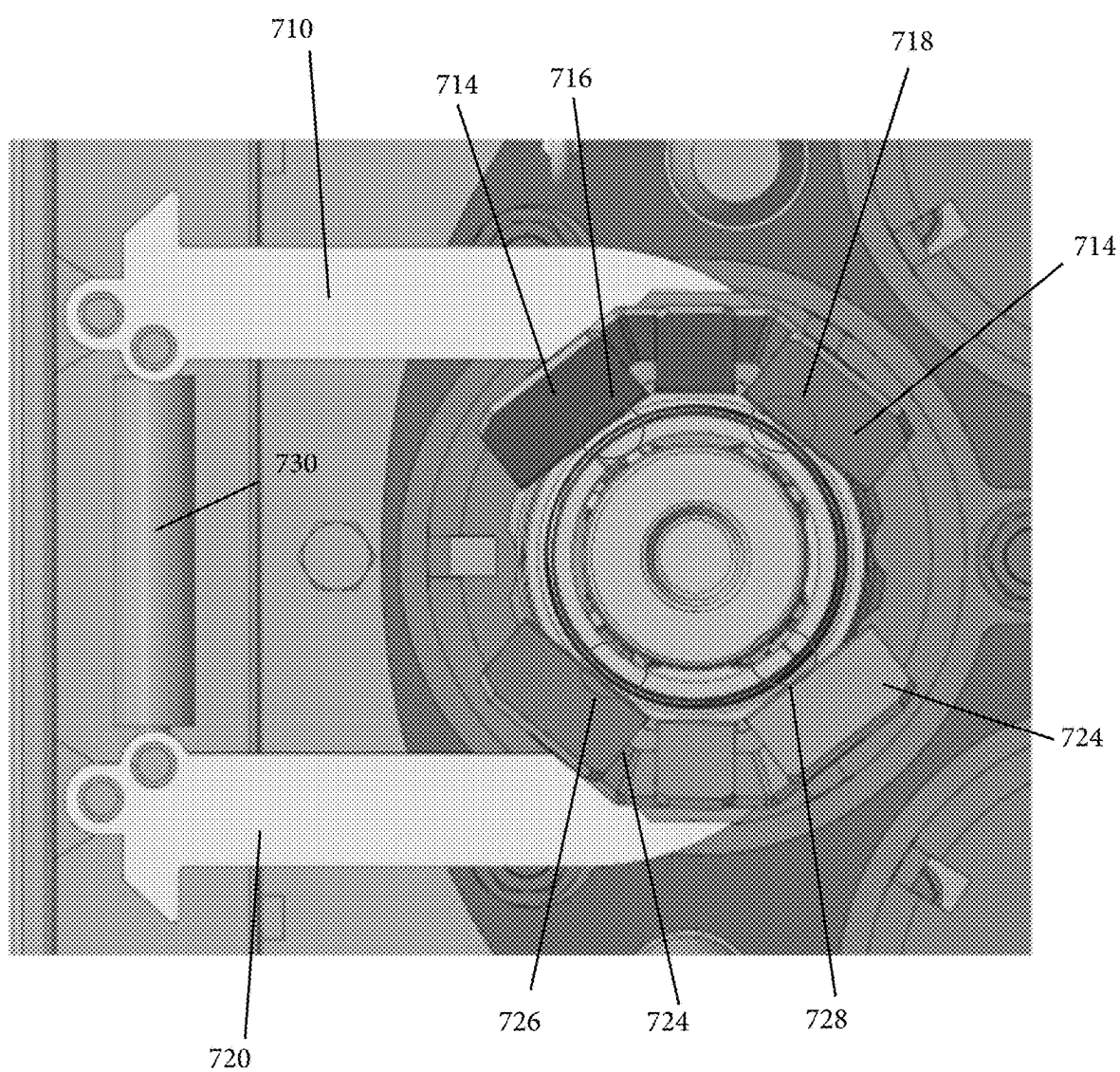


FIG. 23C

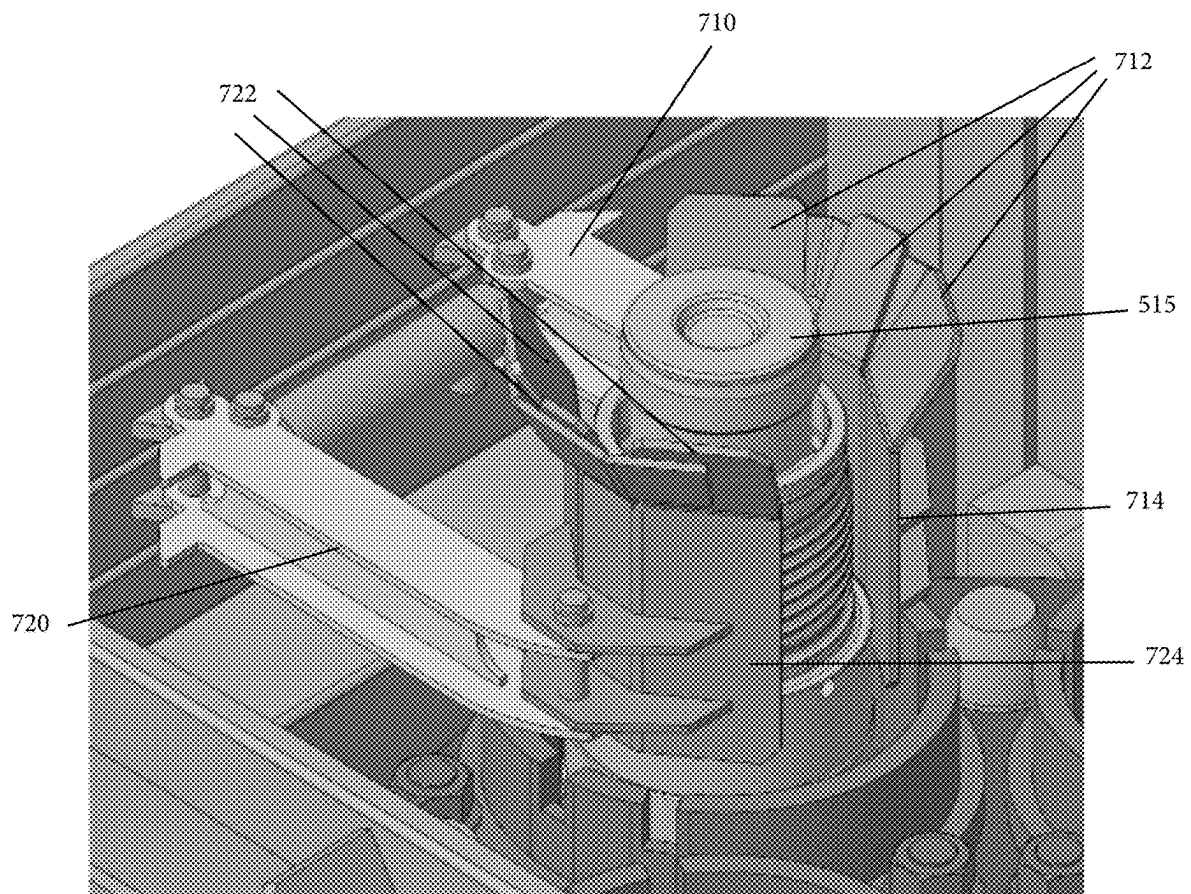


FIG. 24

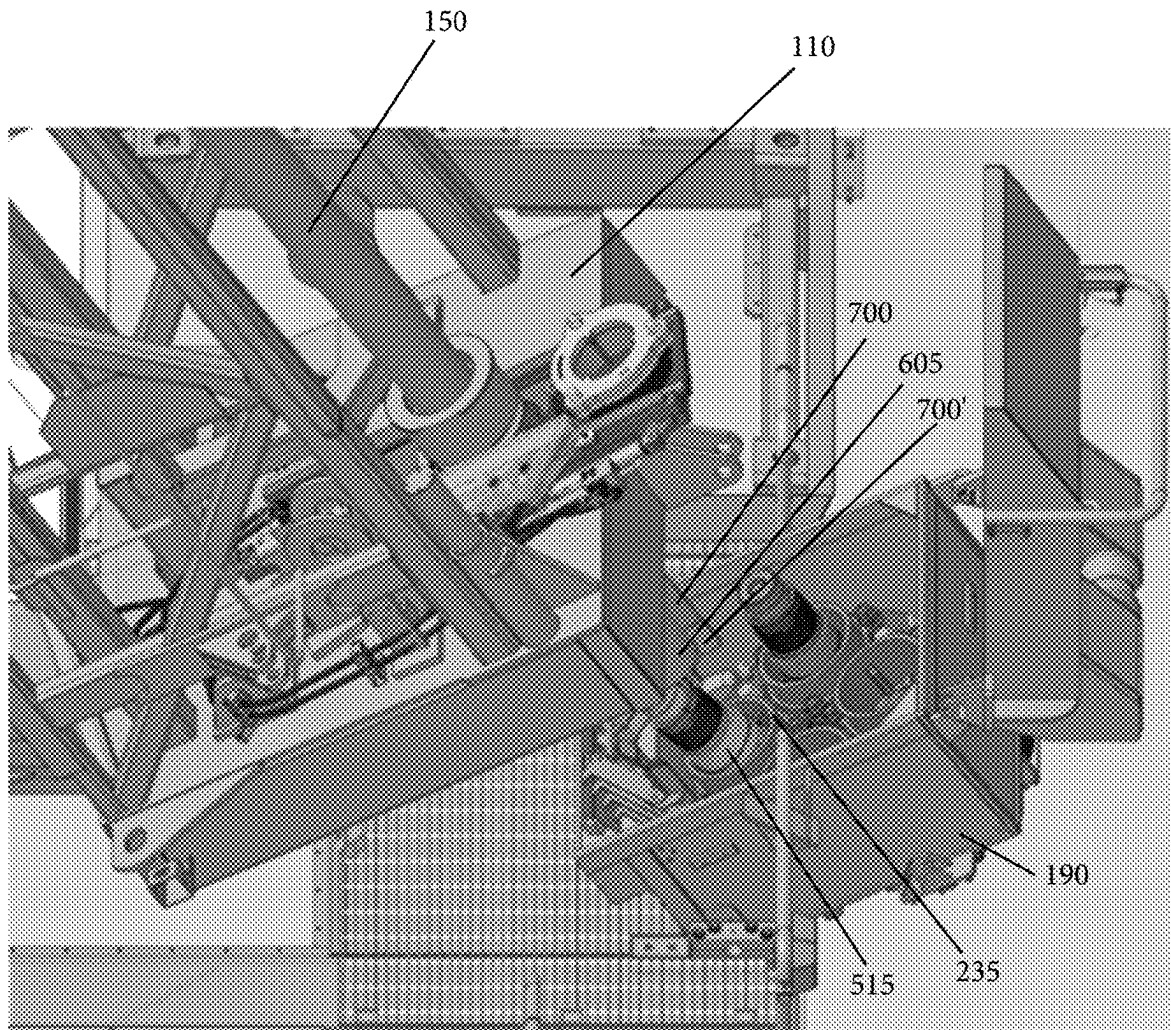


FIG. 25A

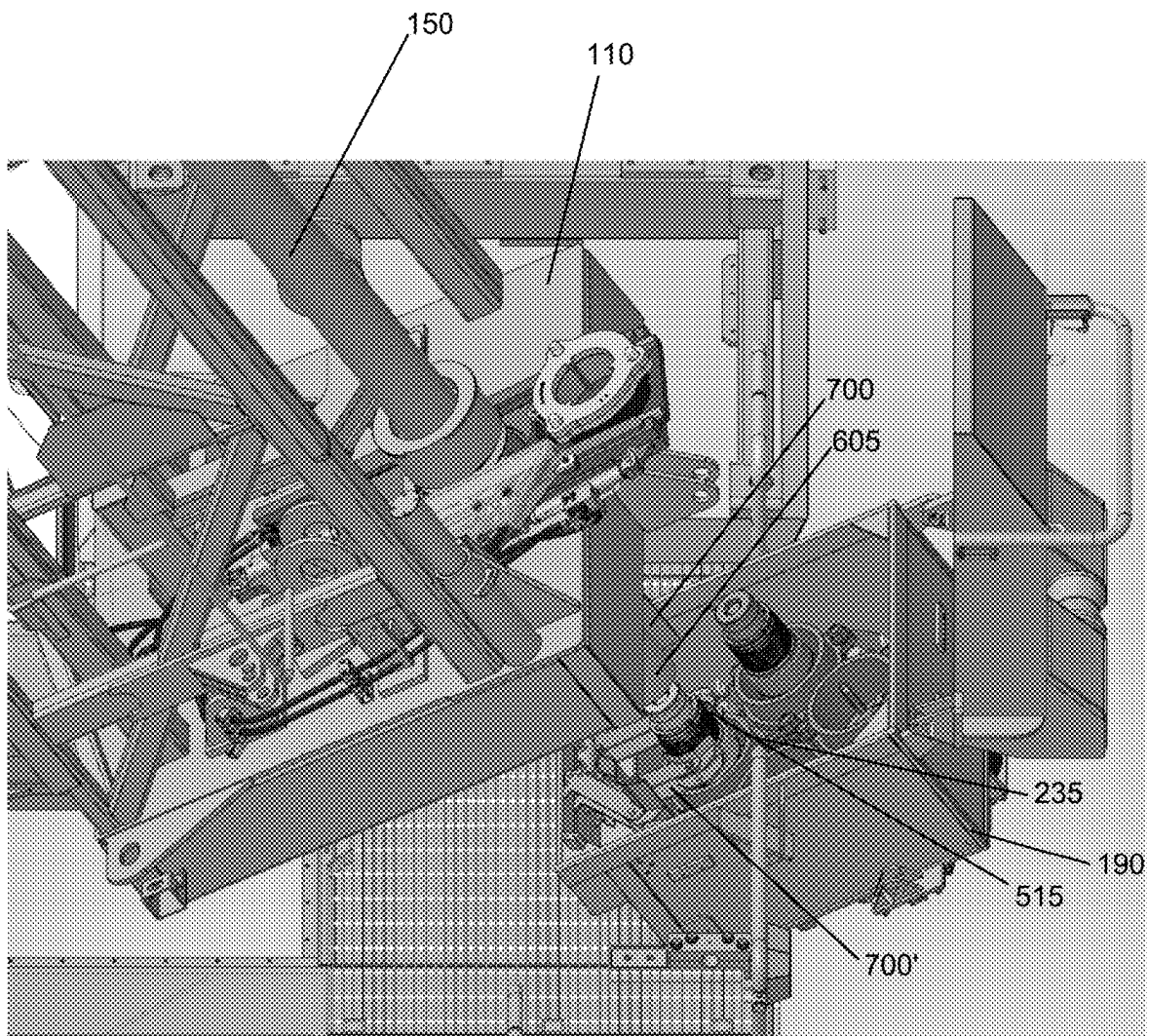


FIG. 25B

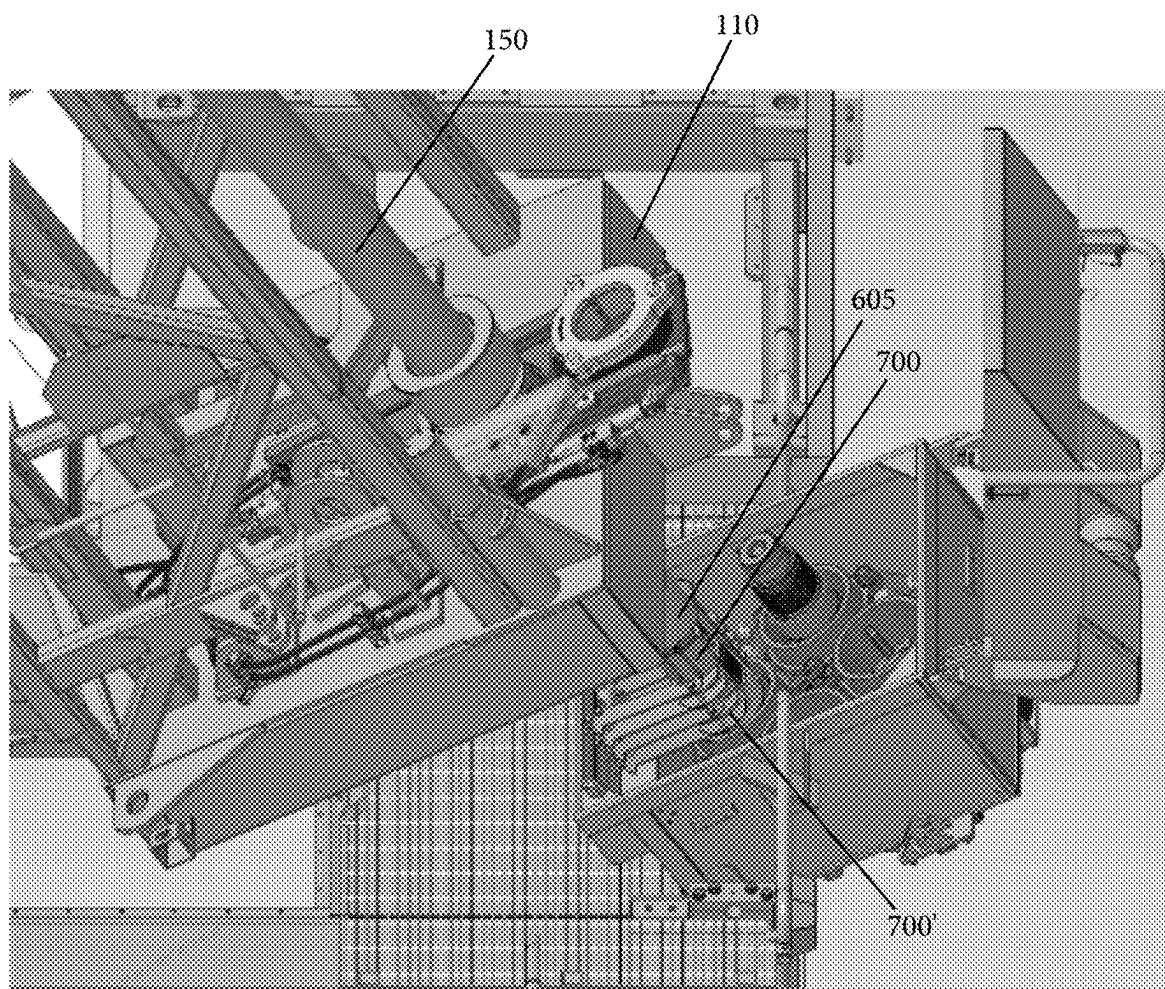


FIG. 25C

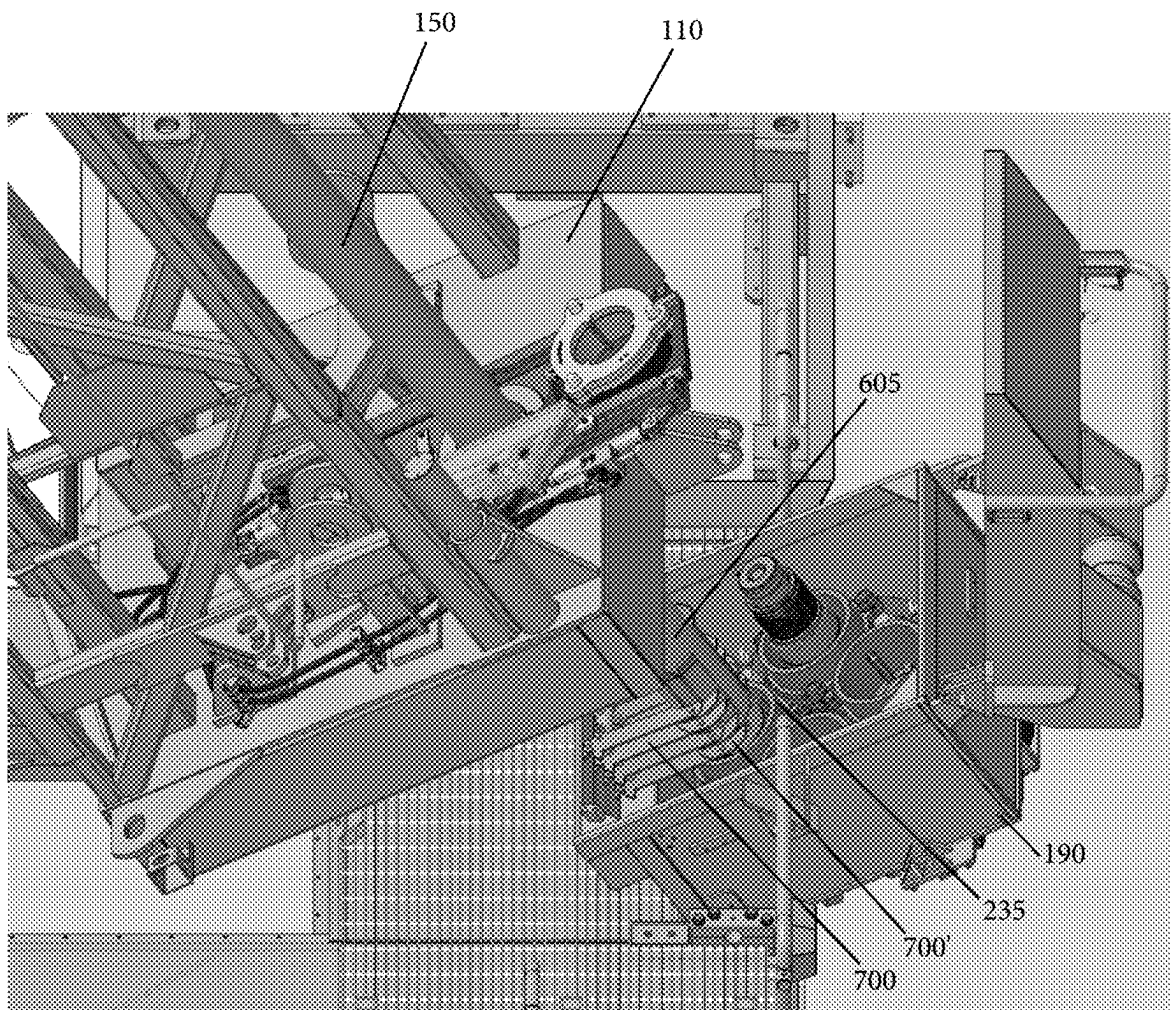


FIG. 25D

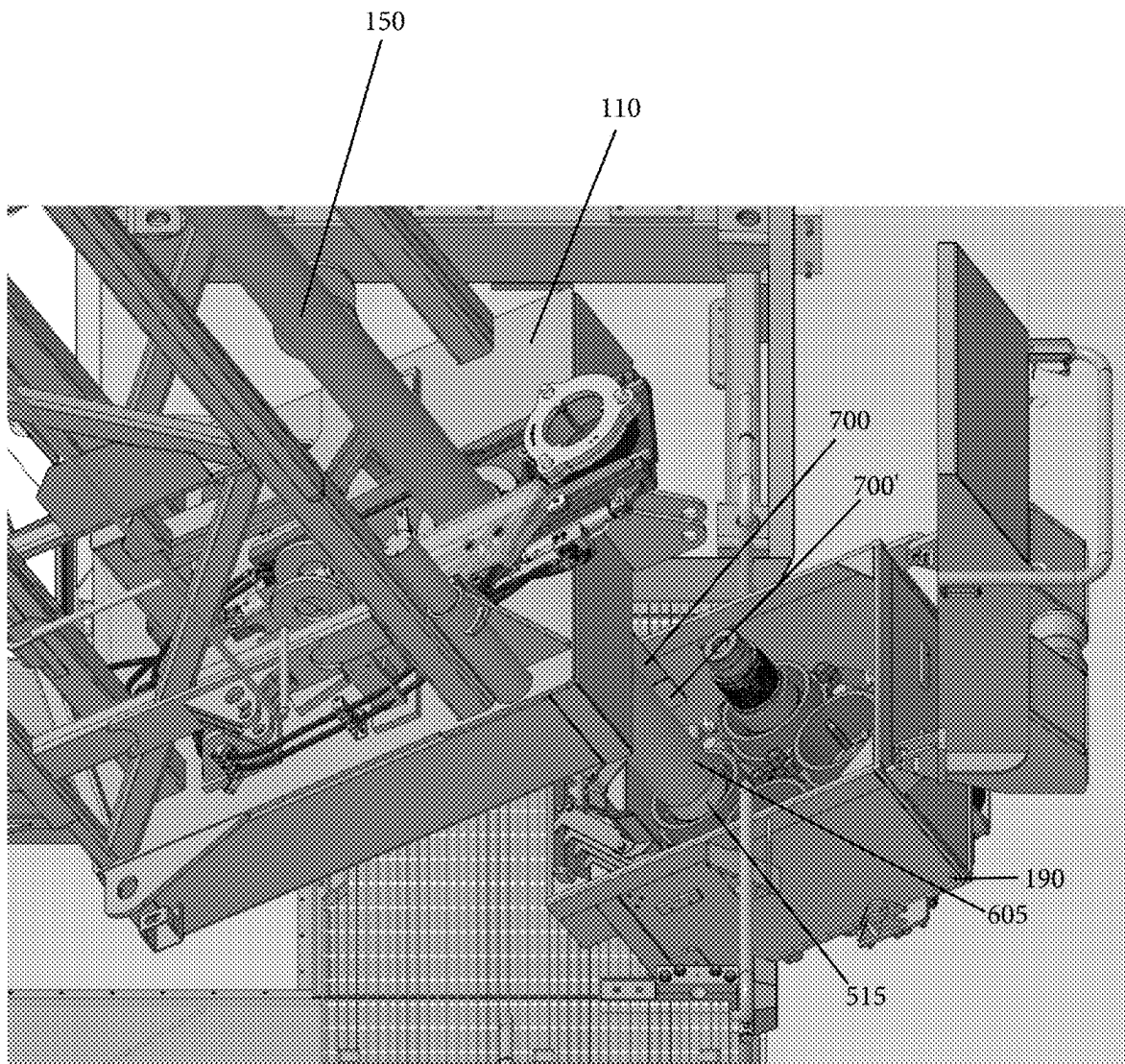


FIG. 25E

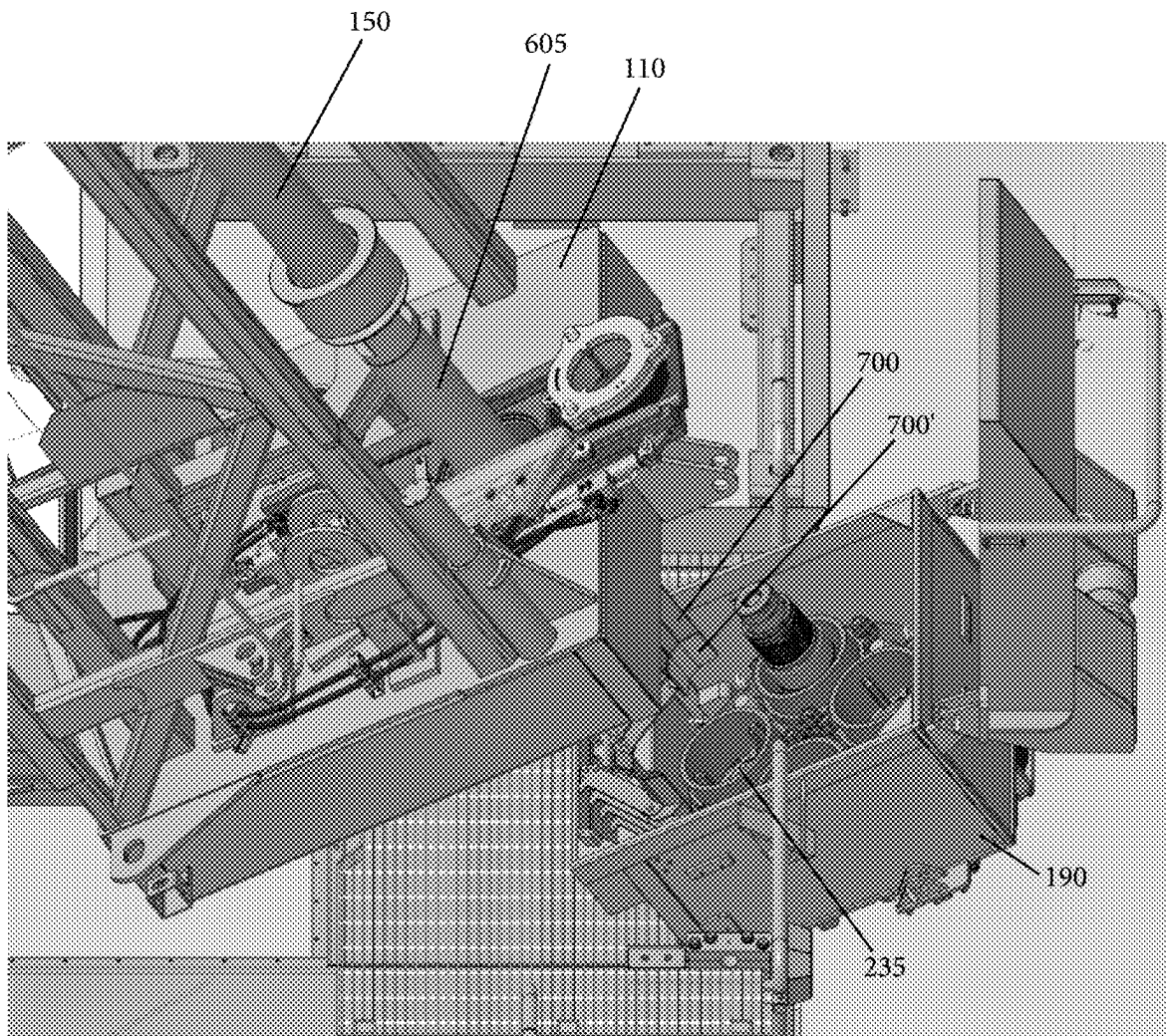


FIG. 25F

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DRILLING TOOL CHANGER APPARATUS**CROSS-REFERENCE TO RELATED APPLICATIONS**

This application is a continuation-in-part of U.S. Non-Provisional patent application Ser. No. 17/899,012 filed Aug. 30, 2022 and entitled “Drilling Tool Changer Apparatus”, now U.S. Pat. No. 11,898,406, which is a continuation of U.S. Non-Provisional patent application Ser. No. 17/161,880 filed Jan. 29, 2021 and entitled “Drilling Tool Changer Apparatus”, now U.S. Pat. No. 11,466,522, each of which is incorporated herein by reference for all purposes.

TECHNICAL FIELD

This disclosure relates to drilling systems and more particularly to a drilling tool changer apparatus for use in applications such as mining, blast-hole drilling and other down-hole drilling applications.

BACKGROUND

This background section provides a context to the invention recited in the claims. The description here may include concepts that could be pursued but are not necessarily ones that have been previously conceived or pursued. Therefore, unless otherwise indicated, what is described in this section is not prior art to the description and claims in this application and is not admitted to be prior art by inclusion in this section.

Drilling systems generally have a vertical drill tower, or mast, constructed from structural members such as steel beams and reinforcing supports. The drill tower is often coupled to a mobile platform, usually self-propelled, for positioning the drill tower in a desired location to conduct a drilling operation. The drill tower is often equipped with a drill magazine or holder of some sort, which is adapted to support a drill string formed from a combination of drill extenders, usually called drill rods or drill pipes. The drill magazine is used to selectively add the drill pipes to the drill string for drilling a hole having a desired depth. The drill magazine is intended to allow a drilling operation to progress into the drill hole by making readily available a continuous string of drill pipes as needed for advancing a drilling tool into a drill hole.

The lowermost drill pipe in the drill string is configured to receive a drilling tool at its lower end to conduct the drilling operation. The drilling tool is usually a drill bit or a down-the-hole (“DTH”) bit and hammer chuck assembly but could be a sub with a drilling tool attached. (In this disclosure, the term “drill bit” may be used interchangeably for the general term “drilling tool”, and the term “drill pipe” may be used interchangeably for the term “drill string”, unless the context requires otherwise.) After a certain amount of usage in the drilling operation it is often desirable or necessary to remove and replace the drilling tool due to accumulated wear of an in-service drill bit, or the need to change between a drill bit and a hammering tool (or vice versa), etc. In order to minimize downtime in the drilling operation due to change-out of drilling tools, mechanisms may be provided to facilitate removing one drilling tool from the end of the drilling string and replacing it with another drilling tool from a storage or supply location.

What is needed is a replaceable magazine or carousel for drilling tools that does not require a large amount of space to accommodate swinging of an arm for the magazine, or for

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rotation of the magazine, which magazine can accommodate selecting drilling tools other than in a linear order, and which is readily adaptable for use in an automated or semi-automated manner to improve access for changing drilling tools, to minimize downtime during change-outs and to minimize the number of personnel required at the drilling rig.

SUMMARY

In accordance with one embodiment, an apparatus for changing a drilling tool for a drilling rig is described, where the drilling rig comprises a drill tower supporting a drill pipe. The apparatus for changing a drilling tool comprises a drilling tool changer assembly moveable between a storage position and an exchange position; where the drilling tool changer assembly supports a rotatable carousel assembly. The rotatable carousel assembly has a plurality of bit adaptors for holding drilling tools and is selectively rotatable to bring a selected drilling tool into coaxial alignment with the drill pipe when the drilling tool changer assembly supporting the rotatable carousel assembly is moved into the exchange position. The apparatus may include means for aligning the selected drilling tool within a bit adaptor holding the selected drilling tool, and/or the apparatus may include means for aligning the drill pipe or a hammer with the selected drilling tool.

This Summary is provided to introduce a selection of concepts in a simplified form that are further described below in the Detailed Description. This Summary is not intended to identify key features or essential features of the claimed subject matter, nor is this Summary intended to be used to limit the scope of the claimed subject matter.

BRIEF DESCRIPTION OF THE DRAWINGS

Non-limiting embodiments of the present disclosure are described by way of example in the following drawings, which are schematic and are not intended to be drawn to scale:

FIG. 1 shows a side view of an embodiment of a typical drilling machine having a tower capable of tilting, where the tower is raised to a vertical position.

FIG. 2 shows schematically a general control system of the drilling machine, where a computer and peripherals are connected to operate the disclosed drilling tool changer apparatus.

FIG. 3 is a perspective view of the rig platform showing a drilling tool changer assembly positioned near the break-out area of the tower.

FIG. 4 is a side view of the structure in FIG. 3 showing the lid to the drilling tool changer assembly enclosure opened.

FIG. 5 is a side view of the structure of FIG. 4, showing the drilling tool changer assembly tilted to engage with a drill pipe in the tower; the drill pipe having a drilling tool attached.

FIG. 6 is a side view of the structure of FIG. 4, showing the drilling tool changer assembly tilted and engaged with a drill pipe for removal or addition of a drilling tool to the drill pipe.

FIG. 7 is a side view of the structure of FIG. 4 showing the drilling tool changer assembly tilted and a drill pipe withdrawn from the assembly after removal of a drilling tool from the drill pipe.

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FIG. 8 is a perspective view of the first embodiment of the carousel of a drilling tool changer assembly including an enclosure, with bit adaptors shown holding a plurality of drilling tools.

FIGS. 9A and 9B comprise perspective views of an embodiment of the carousel of the drilling tool changer assembly, where FIG. 9A is a top view of the assembled carousel shown holding drill bits, and FIG. 9B is a perspective view of an embodiment of the rotatable carousel assembly viewed from the rear.

FIGS. 10A and 10B comprise views of an alignment plate of the rotatable carousel of the drilling tool changer assembly, where FIG. 10A is a perspective view of the alignment plate showing alignment pins to be received by the carousel plate, and FIG. 10B is a plan view of the structure of FIG. 10A, showing the off-center location of one alignment pin.

FIG. 11 is a perspective view of an embodiment of the drilling tool changer assembly, viewed from the underside of its enclosure and illustrating a sensor arrangement for detecting the angular position of the rotatable carousel.

FIGS. 12A and 12B comprise views of the rotatable carousel of the drilling tool changer assembly, where FIG. 12A is a plan view of the rotatable carousel of the drilling tool changer assembly, and FIG. 12B is a perspective view of the carousel without drilling tools in the holders for the same.

FIGS. 13A and 13B comprise perspective views of bit adaptors; FIG. 13A is an adaptor having a welded bottom plate, and FIG. 13B is an adaptor having a removable bottom plate.

FIGS. 14A and 14B comprise perspective views of an embodiment of the bit adaptors for the drilling tool changer. FIG. 14A shows empty bit holders on the carousel of the drilling tool changer assembly; FIG. 14B shows a plurality of bits or drilling tools disposed in bit adaptors of possibly varying sizes on a carousel plate, and FIG. 14B shows a typical bit adaptor.

FIGS. 15A and 15B comprise views of an embodiment of the drilling tool changer assembly, where FIG. 15A is a plan view depicting bit adaptors of various sizes, and FIG. 15B is a side cut-away view of bit adaptors of various sizes on a carousel.

FIG. 16A is a side plan view of a down-the-hole (DTH) tool positioned within a bit adaptor, and FIG. 16B depicts a side plan view of a DTH tool positioned within an embodiment of a pivoting bit adaptor.

FIG. 17A is a top-down perspective view of the pivoting bit adaptor of FIG. 16B in the open position, and FIG. 17B is another top-down perspective view of the pivoting bit adaptor of FIG. 16B in the open position and rotated approximately 90 degrees.

FIGS. 18A, 18B and 18C comprise a sequence of side cross-sectional views of a DTH tool being lowered into the pivoting bit adaptor of FIG. 16B, FIG. 17A and FIG. 17B, where FIG. 18A shows the pivoting bit adaptor in the open position with the DTH tool making initial contact with a shim pad on a pivoting component, FIG. 18B shows the pivoting bit adaptor with the pivoting component in a partially rotated position with the DTH tool partially lowered into the pivoting bit adaptor, and FIG. 18C shows the pivoting bit adaptor with the pivoting component in a fully rotated position with the DTH tool fully lowered into the pivoting bit adaptor.

FIGS. 19A and 19B comprise perspective views of an embodiment of a side swinging drill string support, where FIG. 19A shows the side swinging drill string support in a

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stowed configuration, and FIG. 19B shows the side swinging drill string support in an activated configuration.

FIG. 20A is a perspective view of an embodiment of a side swinging drill string support with a roller sized for a larger diameter drill string. FIG. 20B is a perspective view of another embodiment of a side swinging drill string support with a roller sized for a smaller diameter drill string.

FIGS. 21A and 21B comprise a sequence of side, partial cut-away views of a side swinging drill string support in a representative operational environment, where FIG. 21A shows the side swinging drill string support in the stowed configuration and a hammer or drill pipe on the end of a drill string sagging out of alignment, and FIG. 21B shows the side swinging drill string support in the activated configuration with its roller engaging the hammer or drill pipe to align it with the drilling tool.

FIGS. 22A and 22B comprise perspective views of an embodiment of an alignment clamp, where FIG. 22A shows the alignment clamp in an open position, and FIG. 22B shows the alignment clamp in a closed position.

FIGS. 23A, 23B and 23C comprise a sequence of top plan views of the alignment clamp of FIGS. 22A and 22B in a representative operational environment, where FIG. 23A shows the alignment clamp in an open position and a DTH tool in a tilted position within a bit adaptor, FIG. 23B shows the alignment clamp in a partially closed position, making initial contact with the DTH tool, and FIG. 23C shows the alignment clamp in a closed position with the DTH tool aligned with the bit adaptor.

FIG. 24 is a side perspective view of the structure of FIG. 23C, with the alignment clamp shown in the closed position and with the DTH tool aligned with the bit adaptor.

FIGS. 25A, 25B, 25C, 25D and 25E comprise a sequence of side perspective views of an embodiment of an enclosure that houses lower and upper alignment clamps in a representative operational environment. FIG. 25A shows the lower and upper alignment clamps in open positions, respectively, and a DTH tool tilted within a bit adaptor carried by a carousel assembly within the enclosure. FIG. 25B shows the lower alignment clamp in a closed position around the DTH tool to align the bit within the bit adaptor. FIG. 25C shows the upper alignment clamp in a closed position to guide a hammer on the end of a drill string into alignment with the DTH tool. FIG. 25D shows the lower and upper alignment clamps in closed positions, respectively, to maintain alignment as the hammer is threaded onto the DTH tool. FIG. 25E shows the lower and upper alignment clamps retracted to open positions, respectively, once the joint is made. FIG. 25F shows the lower and upper alignment clamps in open positions, respectively, and the drill string, hammer, and DTH tool retracted out of the enclosure.

DETAILED DESCRIPTION

FIG. 1 shows a side view of a typical mobile drilling machine 100. The drilling machine has a tower 110, a platform 120 supporting the tower 110, and typically tracks 130 or wheels for propelling the drilling machine over ground. FIG. 1 also shows an operator's cab 140 situated on the platform 120. The term "cab" in this disclosure refers to either a housing for an operator or a workstation location on the platform 120, which may or may not be occupied by an operator; the latter would be the case in autonomous machines. FIG. 1 shows the typical engine compartment or "power pack" 105, and mechanical accessories of a drilling machine 100. The tower 110 will also carry a drill string 150 extending through the tower 110, and a rotary head 145,

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being typically a hydraulic or electric motor for rotating the drill string 150. The drill string 150 is terminated at its lower end by a removable bit or drilling tool 215. In general, the tower 110 is pivotably moveable so that its tilt allows operations for the change-out of drilling tools 215. The drill string 150 and its components will be discussed in more detail below. The reader should note that the improvements disclosed here could be embodied in a fixed drilling machine as well as the mobile machine illustrated, and the claims cover both embodiments.

The drilling machine 100 further includes a control system 155, which is operatively coupled to the power pack 105. The associated implements operatively connected to the power pack 105 and the control system 155 are described in more detail below and suggested only schematically in FIG. 1. FIG. 2 shows schematically the control system 155 operatively associated with the example drilling machine which here includes the disclosed drilling tool changer apparatus. The control system 155 comprises generally a computer 160 that is typically a programmable digital computer, further comprising a read-only memory, a non-transitory computer readable storage medium for storing instructions executable by a processor (such as a random-access memory), a central-processing unit or processor, and a hard drive or flash memory or the like for further storage of programs and data, as well as input and output ports. FIG. 2 shows the example control system 155 operatively connecting the computer 160 to a visual display 165 for an operator, and a control input interface 170, such as a joystick, or touch screen, or both. In this disclosure, the term “computer” may be used broadly to refer to both a programmable digital computer as just described, the software executed by the computer, and relevant peripheral devices connected to a computer, as well as networks of computers. One industrial example of a computer 160 suitable for controlling the operations of actuators in a tracked vehicle is the Rig Control System (RCS) provided by Epiroc Drilling Solutions, LLC. The computer 160 connects by means of a bus 115, which may be a wired or wireless network, to the various sensors and actuators shown in FIG. 2 (and in some embodiments, others).

The control system 155 includes one or more control inputs which can be adjusted by the operator in the operator’s cab 140. These may include one or more input controls for controlling the operation of the tower 110, including its tilt angle, and also the operations of the drilling tool changer assembly 190, described below, which is a component of the drilling tool changer apparatus.

FIG. 3 is a perspective view of the rig platform showing the drilling tool changer assembly 190 positioned near the break-out area of the tower. Referring to FIG. 3, we see a drilling tool changer assembly 190, comprising an enclosure 195 for the carousel assembly 230 (see FIGS. 7 and 8). The drilling tool changer assembly 190 is disposed on the rig platform 120 and mounted by a hinge 220 so as to rotate upward from the rear (to the left in FIG. 3) when urged upward by changer actuators 225. FIG. 3 also shows breakout tool 180 adjacent to the base of the tower 110, and the operator’s cab 140 that would typically be situated adjacent to tower 110 on the drilling rig 100. The enclosure 195 has a lid 200 that can be opened by action of one or more lid actuators 210, which may be hydraulic or electric motors or the like. The lid 200 is not required, but is desirable to keep dirt and foreign objects from falling into the drilling tool changer assembly 190.

It is advantageous that the disclosed drilling tool changer assembly 190 is not located over the opening in the drill

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deck 120, where it could interfere with breakout operations, and further, is not located on or connected to the tower 110, where it would add extra weight and complicate balancing the tower 110.

FIG. 4 shows a side view of the structures of FIG. 3, where the lid 200 of the carousel enclosure 195 is opened. In FIG. 5, the drilling tool changer assembly 190, with the lid 200 of the carousel enclosure 195 opened, is tilted forward on the hinge 220 by action of actuators 225, thus moving from a storage position as shown in FIGS. 3 and 4, to an exchange position, as shown in FIGS. 5 and 6. FIG. 6 shows the tilted drilling tool changer assembly 190 receiving a drilling tool 215 from the drill string 150 held by the now-tilted tower 110, where the drilling tool changer assembly 190 is tilted at an angle to substantially match the tilt of the tower 110. FIG. 7 shows a drill string 150 after a drilling tool 215 has been removed and placed in the drilling tool changer assembly 190. FIGS. 5-7 could also illustrate the opposite operation, where a drilling tool 215 is attached to the drill string 150. FIGS. 5-7 show an actuator or motor 280 for rotating the carousel assembly 230. (The actuator or motor 280 may include a gearbox.) The apparatus and procedures for the change-out or replacement of the drilling tool 215 are described in more detail below.

Drilling Tool Changer Carousel Assembly

FIG. 8 is a perspective view of an embodiment of the rotatable carousel assembly 230 of a drilling tool changer assembly 190, situated in an enclosure 195, with bit adaptors 235 shown holding a plurality of drilling tools 215. The rotatable carousel assembly 230 is removable from the drilling tool changer assembly 190 and replaceable therein, as will be described. The rotatable carousel assembly 230 has a plurality of receptacles, called here bit adaptors 235, for receiving drilling tools 215. The rotatable carousel assembly 230 is selectively rotatable, in either direction, via a motor 280, to bring a selected drilling tool 215 into coaxial alignment with the drill string 150 when the drilling tool changer assembly 190 supporting the rotatable carousel assembly 230 is moved into the exchange position as shown in FIGS. 5-7. FIG. 8 also shows an empty bit adaptor 235 in a pre-determined angular position 420 of the rotatable carousel assembly 230, where a used drilling tool 215 may be deposited after removal from the drill string 150. Alternatively, this pre-determined angular position 420 could be the current position of a drilling tool 215 held in the bit adaptor 235, to be attached to the drill string 150. In either case, the rotatable carousel assembly 230 will be rotated to bring the relevant bit adaptor 235 into the correct pre-determined angular position 420 for the pick-up or drop-off of a drilling tool 215 as determined by an operator or the control system 155.

The rotatable carousel assembly 230 has at least one alignment pin 245 situated on the alignment plate 255 below a carousel plate 285, which alignment pin 245 passes through a slot, notch, or hole 290 in the carousel plate 285, so that the carousel plate 285 can be held on the alignment plate 255 in a fixed angular position within the rotatable carousel assembly 230. The carousel plate 285 of this embodiment further comprises at least one off-center alignment hole 295, comprising a slot, notch or hole, mating with an off-center alignment pin 250, for re-aligning the rotatable carousel assembly 230 in the same angular position on the alignment plate 255 after the rotatable carousel assembly 230 is removed from the drilling tool changer assembly 190 and replaced in the drilling tool changer assembly 190. In

this way the rig operator, or an automatic process in a control system 155, can know which drilling tool 215 is in which bit adaptor 235 in the rotatable carousel assembly 230. The feature allows the operator or control system 155 to confirm or select the correct drilling tool 215 for the desired drilling task, because in general, drilling tools 215 may be of different sizes, or types, or even have different conditions of wear, and it is highly advantageous to know without manual inspection which drilling tool 215 is in which position in the rotatable carousel assembly 230, so that drilling operations are not delayed after a change-out of drilling tools 215.

The reader should note that the alignment pin or pins 245, and the off-center alignment pin 250 are shown schematically in the figures, and represent, in general, an indexing feature. These may be any sort of finger, rod, or lug that can engage slots, notches or holes in the carousel plate 285 to maintain the same in a given angular position in the rotatable carousel assembly 230, and thus the terms "alignment pin" or "off-center alignment pin" should be construed broadly according to this indexing function. In particular, in some embodiments therefore, alignment pins 245 or off-center alignment pin 250 do not pierce the carousel plate 285 completely in order to engage it.

In some embodiments, the rotatable carousel assembly 230 may optionally be a first rotatable carousel assembly 230, and the rotatable carousel assembly 230 replaced in the drilling tool changer assembly 190 may be a second rotatable carousel assembly 230, the second rotatable carousel 230 having possibly replaced or exchanged drilling tools 215.

FIG. 9A shows examples of drilling tools 215 resting in the bit adaptors 235. Each bit adaptor 235 is connected to a carousel plate 285. FIG. 9A shows alignment holes 290 in the carousel plate 285 for selectively receiving an alignment pin 245, and also shows at least one off-center alignment hole 295 and off-center alignment pin 250, to allow indexing of the carousel plate 285. The carousel plate 285 may be a single plate, or constructed of two plates bolted together by bolts or retaining pins 240, for ease of manufacturing.

FIG. 9A shows the rotatable carousel assembly 230 as described in above, but including the alignment plate 255 that supports the alignment pins 245 and off-center alignment pin or pins 250. (Generally, only one off-center pin 250 would be required for alignment, but more than one could be provided, and reference here to an "alignment pin" includes one or more.) The carousel plate 285 and the bit adaptors 235 are preferably bolted together with bolts 240. FIG. 9A shows the alignment plate 255 supporting the alignment pins 245 and off-center alignment pin 250, rotatable by a bull gear 260. As shown in FIG. 9A, the carousel plate 285 is removably supported by the alignment plate 255, and the alignment pins 245 and off-center alignment pin 250 thus pass through holes or slots 290 in the carousel plate 285. One of the holes or slots is an off-center alignment hole 295, so that the carousel plate 285 (and thus the bit adaptors 235) can only be indexed to and replaced in one position on the alignment plate 255.

FIG. 9B is a perspective view of this embodiment of the rotatable carousel assembly 230 viewed from the rear or bottom of the assembly depicted in FIG. 9A. FIG. 9B shows how the carousel plate 285 is supported on the alignment plate 255, and how the alignment plate 255 is fastened to a bull gear 260, which bull gear 260 rotates the alignment plate 255, and thus the carousel plate 285, when rotation is imparted to the bull gear 260 by an actuator, such as a motor 280 and pinion gear (not shown in FIG. 9B). The rotation imparted may be in either direction, under command of an

operator or an automatic process in a control system 155, to index the rotatable carousel assembly 230 to the desired position, as explained below.

FIG. 10A show a perspective view of the alignment plate 255 supported by the bull gear 260. The alignment plate 255 supports a plurality of alignment pins 245, and at least one off-center alignment pin 250. As described above with reference to FIGS. 9A and 9B, the carousel plate 285 is supported by the alignment plate 255, where alignment holes 290 (and off-center alignment hole 295) in the carousel plate 285 receive, respectively, alignment pins 245 and the at-least one off-center alignment pin 250. This structure is shown in plan view in FIG. 10B, where the circumference 300 of a circle on the radius of the alignment pins 245 is shown, illustrating the off-center location of the at least one off-center alignment pin 250, as well as the respective alignment holes 290 and off-center hole 295 in the carousel plate.

FIG. 11 illustrates an embodiment of a sensor arrangement for detecting the angular position of the rotatable carousel assembly 230. A magnet 370 or equivalent is connected to a pin or shaft 390, which pin or shaft 390 is connected to the alignment plate 255 and compelled to rotate therewith. A sensor 380 detecting the position of the magnet 370 is fixed to a bracket 400 or other convenient structure that is fixed to the enclosure 195 of the drilling tool changer assembly 190, so that as the alignment plate 255 rotates by action of the motor 280, the change in output from the sensor 380 corresponds to the angular position of the alignment plate 255, and thus the angular position of the rotatable carousel assembly 230, which is rotated by action of the alignment plate 255. Because the rotatable carousel assembly 230 only fits the alignment plate 255 in one predetermined position, the output of the sensor 380 uniquely reports the angular position of the rotatable carousel assembly 230, which output is preferably communicated to the control system 155 by means of a connector 410, which can be a cable or wireless connection. A suitable magnetic sensor of this type would be the SMART Position Sensor configuration, manufactured by Honeywell International, Inc., or equivalent.

FIGS. 12A and 12B are a more detailed view of the carousel assembly 230 of the drilling tool changer assembly 190, FIG. 12A being a plan view and FIG. 12B being a perspective view of the same. In FIG. 12A, three bit adaptors 235 are occupied by drilling tools 215, and one bit adaptor 235 is empty. (The reader should note that the number of bit adaptors 235 in a carousel assembly may be more or fewer than the four shown.) In FIG. 12B, for clarity, all the bit adaptors 235 are shown as empty.

FIGS. 12A and 12B show a carousel plate 285, which may be one integral plate, or, for convenience of construction, two or more plates fastened together, with bolts, such as the retaining pins or bolts 240 shown. The carousel plate 285 supports bit adaptors 235 (described in more detail below), for receiving drilling tools 215. The carousel plate 285 has holes 290 for receiving alignment pins 245, and at least one off-center hole 295 for receiving an at least one off-center alignment pin 250. In this way, a particular type of drilling tool 215 located in a particular bit adaptor 235, and be replaced in the same position in the carousel assembly 230, if the carousel assembly 230 is removed and replaced in the drilling tool changer assembly 190.

Further referring to FIGS. 12A and 12B, the figures show lifting rings 270, where by a lifting apparatus, such as a crane, can attach to and lift out the carousel assembly 230 for replacement of drilling tools 215 in the bit adaptors 235. The

bit adaptors **235** are shown fastened to the carousel plate **285** by retaining pins or bolts **240**. As illustrated below, each bit adaptor **235** further comprises a breaker plate **310** and a bit holder **320**. Alternatively, the breaker plate **310** may be fastened to a bit holder plate **330** that may be removable. The breaker plate **310** of the bit adaptors **235** comprises nubs **275** in its inner circumference to hold the drilling tool **215** and resist the torque of the drill pipe **205** when the drill pipe **205** is rotated by the break-out tool mechanism **180**, as a drilling tool **215** is added to or removed from the drill pipe **205**. Therefore, no special sub or supplemental adaptor to the drilling tool **215** is required for the breakout operations.

FIGS. **13A** and **13B** show more detail of bit adaptor **235** of the drilling tool changer assembly **190**. In FIG. **13A**, an embodiment has a breaker plate **310** with nubs **275**, where the nubs **275** are sized to grasp a drilling tool **215** of a pre-determined size and restrain its movement against the torque of the break-out operation. In FIG. **13A**, the bit holder **320** and the bit holder plate **330** are alternatively welded together to form an integral unit. Mounting holes **265** are provided for mounting the bit adaptor **235** to the carousel plate **285**. In the embodiment of FIG. **13B**, the breaker plate **310** is connected to a bit holder plate **330**, where the bit holder plate **330** supports a drilling tool inserted into the bit adaptor **235**. The embodiment of FIG. **13A**, providing a more shallow receptacle, is adapted to support and hold smaller drilling tools **215** than the embodiment of FIG. **13B**. In either embodiment, the bit adaptor functions to provide storage for the drilling tool **215** in the carousel assembly **230**, while also holding the drilling tool **215** firmly against the torque of the break-out operation.

FIGS. **14A** and **14B** show embodiments of the drilling tool changer assembly **190** where the bit adaptors **235** situated in the carousel assembly **230** are sized to receive drilling tools **215** of differing sizes, both in diameter or in depth, or both dimensions. FIG. **15A** is a plan view of such an embodiment. FIG. **15B** is a side view omitting depiction of the carousel plate **285** for clarity, and showing only the exemplary bit adaptors **235** of varying sizes. In FIG. **15B**, the rightmost bit adaptor **235** depicted has a bit holder plate **330**, but alternatively lacks a bit holder **320**.

Detection of Drilling Tools in Changer Assembly

It is advantageous for an operator or an automatic process to detect if a bit or drilling tool **215** is present in the rotatable carousel assembly **230**, so that no attempt will be made to engage with the drill pipe **205** if no bit is present. Also, it is advantageous that an operator or automatic process can detect which type of bit or drilling tool **215** is actually present in a particular bit adaptor **235**. To that end, embodiments can be provided with these capabilities. Radio-frequency identification (RFID) technology, for example, is known in the art. FIG. **8** illustrates an RFID sensor **340** situated at some convenient location in the enclosure **195** of the rotatable carousel assembly **230**. The RFID sensor **340** can read a corresponding RFID tag (not shown) on a drilling tool **215** and transmit this identification to an operator or automatic process in the control system **155**, where it may be stored for further reference or use in the automatic process. For example, an automatic process could select the desired size and type of drilling tool **215** for particular rock strata or drilling conditions. Equivalently, electromagnetic sensors could be employed, such as near-field communications devices operating on principles generally within RFID technology, as is known in the art. In other embodiments, the

RFID sensor **340** could be a plurality of RFID sensors **340**, where each is associated with a particular bit adaptor **235**.

Further, FIG. **8** shows an ultrasonic detector **350**, which provides the capability of detecting whether or not a drilling tool **215** is present in a particular bit adaptor **235** of the rotatable carousel assembly **230**. In the case of both the RFID sensor **340** and the ultrasonic detector **350**, the relevant index or angular position of the rotatable carousel assembly **230** (as known by the means described in connection with FIG. **11** above) can be correlated with the identification returned from the RFID sensor **340**, or with the signal from the ultrasonic detector **350** to provide this information to an operator or to an automatic process in the control system **155**, where it may be stored for future reference or use in an automatic process. Equivalents in other embodiments could be a radar sensor to detect presence of a bit **215**, or a switch.

Down-the-hole (DTH) Tool and Hammer Alignment

In some implementations, the drilling tool **215** may be a down-the-hole (“DTH”) drill bit and hammer chuck assembly (collectively referred to herein as a “DTH tool”), and the drill pipe **205** at the end of the drill string **150** may be replaced with a hammer that includes receiving threads to mate with the threads of the DTH tool. The DTH tool threads and the corresponding hammer threads are less tapered and therefore tend to be more susceptible to cross threading than the API or BECO threads on standard rotary drill bits and drill pipes. As such, a more precise alignment between the DTH tool and hammer may be required to make the joint and avoid cross-threading.

However, because the DTH tool includes both a hammer bit and a hammer chuck, the DTH tool is relatively longer and more top heavy than standard rotary drill bits such that when the drilling tool changer assembly **190** is tilted to the exchange position shown in FIGS. **5-7**, the DTH tool is more likely to tilt within the bit adaptor **235** rather than staying aligned with the drill string **150**.

FIG. **16A** depicts a side plan view of a DTH tool **515** positioned within the previously described bit adaptor **235** of the present disclosure, and FIG. **16B** depicts a side plan view of a DTH tool **515** positioned within an embodiment of a pivoting bit adaptor **535** of the present disclosure. As shown in FIG. **16A**, the top heavy DTH tool **515** tends to tilt over and fall out of alignment in the bit adaptor **235** when the drilling tool changer assembly **190** is in the exchange position, whereas the DTH tool **515** is held in an aligned position in the pivoting bit adaptor **535** when the drilling tool changer assembly **190** is in the exchange position.

FIG. **17A** depicts a top-down perspective view of the pivoting bit adaptor **535** of the present disclosure, and FIG. **17B** depicts another top-down perspective view of the pivoting bit adaptor **535** rotated approximately 90 degrees. The pivoting bit adaptor **535** comprises a bit holder **520** with a bit holder plate **530** coupled to a bottom surface thereof to support a drilling tool **215**, such as the DTH tool **515** positioned within the pivoting bit adaptor **535**. A mounting ring **525** surrounds the bit holder **520** and includes mounting holes **565** for mounting the pivoting bit adaptor **535** to the carousel plate **285**. The pivoting bit adaptor **535** further comprises a pivoting component **510** that includes a semicircular support **508** and a shim support **512**. The shim support **512** is coupled to the semicircular support **508** and extends substantially perpendicular thereto. The shim support **512** forms a trough **513** into which a shim pad **514** is

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positioned. The shim pad 514 can be provided in different sizes for various types and sizes of bits.

The pivoting component 510 is rotationally coupled to the bit holder 520 via a pin/flange assembly 540, and a return spring 545 ensures the pivoting component 510 remains in the open position shown in FIG. 17A and FIG. 17B when the pivoting bit adaptor 535 is empty. As best shown in FIG. 17B, one or more keyed inserts 522 may be provided on an inside wall of the bit holder 520, and these keyed inserts 522 can be provided in different sizes for various types and sizes of bits to inhibit rotation of the drilling tool 215, such as the DTH tool 515, when making or breaking a joint.

FIG. 18A through FIG. 18C depict a sequence of side cross-sectional views of a DTH tool 515 being lowered into a pivoting bit adaptor 535 of the present disclosure. Although the pivoting bit adaptor 535 is shown in a horizontal position in this sequence, the pivoting bit adaptor 535 would perform in the same manner in a tilted position, such as the exchange position shown in FIGS. 5-7.

Referring to FIG. 18A, the pivoting bit adaptor 535 is shown with the pivoting component 510 in the open position, and with the DTH tool 515 making initial contact with the shim pad 514. Referring to FIG. 18B, as the DTH tool 515 continues to be lowered into the pivoting bit adaptor 535, the DTH tool 515 pushes against the shim pad 514 until the force of the return spring 545 is overcome and the pivoting component 510 rotates about pin/flange assembly 540. Referring to FIG. 18C, when the DTH tool 515 is fully lowered into engagement with the bit holder plate 530, the shim pad 514, as well as the keyed insert(s) 522, keep the DTH tool 515 snug within the pivoting bit adaptor 535. The shim pad 514 and keyed insert(s) 522 are also sized to facilitate proper lowering of the pivoting component 510 to the position shown in FIG. 18C. When the pivoting component 510 rotates to the position shown in FIG. 18C, the semicircular support 508 extends horizontally/radially to surround a portion of the DTH tool 515 positioned within the pivoting bit adaptor 535, and the shim support 512 extends vertically/axially within the pivoting bit adaptor 535, with the shim pad 514 engaging the DTH tool 515.

Another alignment issue that may occur when the drill string 150 includes a hammer or a heavier drill pipe 205 at the end is sagging of the drill string 150 out of alignment with the drilling tool 215 when the tower 110 is tilted as shown in FIGS. 5-7.

Referring now to FIG. 19A and FIG. 19B, an embodiment of a side swinging drill string support 600 is depicted in a stowed configuration and in an activated configuration, respectively. The side swinging drill string support 600 is designed to support a hammer 605 or drill pipe 205 at the end of the drill string 150 when making a joint between the hammer 605 or drill pipe 205 and a drilling tool 215 in applications where the tower 110 is tilted and the hammer 605 or drill pipe 205 tends to sag out of alignment with the bit adaptor 235, 535.

The side swinging drill string support 600 includes a mounting flange 605, a piston 610, a bracket 615, and a roller 620. The roller 620 is mounted to the bracket 615 via first support 617 and second support 619. A spindle 618 extends between the supports 617, 619 and through the roller 620 to enable the roller 620 to rotate with respect to the supports 617, 619. The roller 620 and bracket 615 combination are further rotationally coupled to the piston 610 via a pinned connection 625 between the piston 610 and the second support 619.

In the stowed configuration shown in FIG. 19A, the piston 610 is in an extended position. When the piston 610 extends,

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it applies a pushing force to the second support 619 through the pinned connection 625, thereby rotating the bracket 615 and the roller 620 into axial alignment with the mounting flange 605. In the activated configuration shown in FIG. 19B, the piston 610 is in a retracted position. When the piston 610 retracts, it applies a pulling force to the second support 619 through the pinned connection 625, thereby rotating the bracket 615 and the roller 620 to a position that is perpendicular to the mounting flange 605, as shown.

FIG. 20A and FIG. 20B depict implementations of side swinging drill string supports 600, 600' in the activated configuration. The side swinging drill string supports 600, 600' may be identical in most respects, except for the rollers 620, 620', which are sized to support different sized hammers 605/drill pipes 205.

FIG. 21A and FIG. 21B depict the side swinging drill string support 600 in a representative operational environment to provide support to a sagging hammer 605/drill pipe 205 at the end of a drill string 150 and align it with the drilling tool 215, such as the DTH tool 515. As shown in FIG. 21A, the side swinging drill string support 600 is in the stowed configuration, and the centerline 602 of the hammer 605 is sagging out of alignment with the centerline 517 of the DTH tool 515. As shown in FIG. 21B, when the side swinging drill string support 600 is in the activated configuration, the roller 620 of the side swinging drill string support 600 engages the hammer 605 and provides support to maintain alignment between the centerline 602 of the hammer 605 and the centerline 517 of the DTH tool 515.

In various embodiments, the side swinging drill string support 600, 600' may be configured to be operated manually, electrically, pneumatically, hydraulically, or by other methods. In some embodiments, the side swinging drill string support 600, 600' may be operatively coupled to the control system 155 such that the rig operator, or an automatic process in the control system 155, determines when to move the side swinging drill string support 600, 600' from the stowed configuration to the activated configuration and vice versa.

Referring now to FIG. 22A through FIG. 25F, in some embodiments, instead of the pivoting bit basket 535 and/or the side swinging drill string support 600, 600', one or more alignment clamps may be installed within the enclosure 195 for the carousel assembly 230 of the drilling tool changer assembly 190. The one or more alignment clamps may be used to align a drilling tool 215, such as a DTH tool 515, within a bit adaptor 235 and/or to align a drill pipe 205/hammer 605 with the drilling tool 215/DTH tool 515 to make the joint and avoid cross-threading.

FIG. 22A and FIG. 22B depict side perspective views of an embodiment of an alignment clamp 700 shown in an open position and in a closed position, respectively. The alignment clamp 700 comprises a first arm 710, a second arm 720, an actuating piston 730, and mounting connectors 736, 738. The first arm 710 and the second arm 720 are pivotally coupled via pin connections 732, 734 to opposite ends of the actuating piston 730. Each of the first arm 710 and the second arm 720 further comprises a pad system 715, 725, respectively, coupled to the end of the arm 710, 720 opposite the pin connections 732, 734.

Each of the pad systems 715, 725 comprises a plurality of contact pads 714, 724, respectively, that are disposed substantially perpendicular to the plane of the arm 710, 720 and positioned at different angles to one another. The contact pads 714 of the first arm 710 may be angularly oriented as mirror images to the contact pads 724 of the second arm 720. As the alignment clamp 700 moves to the closed position

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shown in FIG. 22B, the plurality of contact pads 714, 724 partially surround and engage the DTH tool 515 to act as a wedge for alignment purposes, as further described herein.

Each of the pad systems 715, 725 further comprises a plurality of flared flanges 712, 722, respectively, each coupled to an upper end of a corresponding contact pad 714, 724 and extending outwardly at an angle therefrom. The flared flanges 712 of the first arm 710 may be oriented as mirror images to the flared flanges 722 of the second arm 720. When the alignment clamp 700 is in the closed position shown in FIG. 22B, the plurality of flared pads 712, 722 form a generally funnel shape operable to guide the hammer 605 for alignment purposes to allow the joint to be made, as further described herein.

FIG. 23A through FIG. 23C depict the alignment clamp 700 of the present disclosure in a representative operational environment. As shown, the alignment clamp 700 may be installed, such as via mounting connectors 736, 738, within an enclosure 195 for a carousel assembly 230 comprising a plurality of bit adaptors 235, one of which contains a DTH tool 515. FIG. 23A through FIG. 23C depict a sequence of top plan views showing the alignment clamp 700 of the present disclosure moving the DTH tool 515 from a tilted position within a bit adaptor 235 to an aligned position within the bit adaptor 235 while the drilling tool changer assembly 190 is tilted to the exchange position shown in FIGS. 5-7.

As the enclosure 195 is tilted to the exchange position, the DTH tool 515 falls out of alignment with the bit adaptor 235. FIG. 23A shows the alignment clamp 700 of the present disclosure in an open position, and the DTH tool 515 in a tilted position within the bit adaptor 235. In this configuration, the actuating piston 730 of the alignment clamp 700 is extended, and the arms 710, 720 of the alignment clamp 700 are pivoted open to allow rotation of the carousel assembly 230 without interference.

FIG. 23B shows the alignment clamp 700 with the arms 710, 720 in a partially closed position, making contact with the DTH tool 515. In more detail, as the actuating piston 730 of the alignment clamp 700 is retracted to move the alignment clamp 700 to a closed position, the arms 710, 720 pivot about their respective pin connections 732, 734 toward the DTH tool 515 until one of the contact pads 714 of the first arm 710 engages the DTH tool 515 at a first contact point 716 and the opposing one of the contact pads 724 of the second arm 720 engages the DTH tool 515 at a second contact point 726. The contact points 716, 726 are roughly 90 degrees offset from one another.

FIG. 23C shows the alignment clamp 700 of the present disclosure in a closed position around the DTH tool 515, which is now aligned with the bit adaptor 235. In more detail, as the actuating piston 730 continues retracting to close the alignment clamp 700, the arms 710, 720 continue to pivot inwardly about their respective pin connections 732, 734, and the contact pads 714, 724 act as a wedge to push the DTH tool 515. As the alignment clamp 700 tightens its grip, the DTH tool 515 is drawn towards a substantially centered position relative to the contact pads 714, 724. In the aligned position shown in FIG. 23C, the contact pads 714, 724 engage the DTH tool 515 at four points of contact 716, 718, 726, 728 that are spaced apart by roughly 90 degrees.

FIG. 24 depicts a side perspective view of the assembly shown in FIG. 23C, with the alignment clamp 700 of the present disclosure shown in the closed position and with the DTH tool 515 aligned with the bit adaptor 235. With the contact pads 714, 724 holding the DTH tool 515 in alignment, the flared flanges 712, 722 form a generally funnel

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shape that will help guide the hammer 605 to maintain alignment between the centerline 602 of the hammer 605 and the centerline 517 of the DTH tool 515 to facilitate making the joint without cross-threading.

In some embodiments, multiple alignment clamps may be used instead of a single alignment clamp. In such embodiments, a lower alignment clamp may align the DTH tool 515 with the bit adaptor 235, and an upper alignment clamp may align the hammer 605 with the DTH tool 515 to avoid cross threading when the joint is made.

FIG. 25A through FIG. 25F depict an assembly using multiple alignment clamps of the present disclosure in a representative operational environment. In one embodiment, a lower alignment clamp 700' is provided that is structurally similar to the alignment clamp 700 of FIG. 22A through FIG. 24, except the lower alignment clamp 700' does not include the flared flanges 712, 722 at the top of the contact pads 714, 724. An upper alignment clamp 700 is also provided that includes the flared flanges 712, 722 at the top of the contact pads 714, 724.

FIG. 25A through FIG. 25F depict a sequence of side perspective views showing a tower 110 carrying a drill string 150 with a hammer 605 at an end thereof and a drilling tool changer assembly 190 tilted to the exchange position. The drilling tool changer assembly 190 includes an enclosure 195 that houses the lower and upper alignment clamps 700', 700, and a carousel assembly 230 with a plurality of bit adaptors 235, one of which holds a DTH tool 515.

FIG. 25A shows the lower and upper alignment clamps 700', 700 in open positions, respectively, and the top heavy DTH tool 515 tilted within the bit adaptor 235. FIG. 25B shows the lower alignment clamp 700' in the closed position around the DTH tool 515 to thereby align the DTH tool 515 within the bit adaptor 235 to allow for threading to the hammer 605.

Once the DTH tool 515 has been aligned, the drill string 150 can then move the hammer 605 toward the DTH tool 515. During this movement, as shown in FIG. 25C, the upper alignment clamp 700 is closed, and the flared flanges 712, 722 of the upper alignment clamp 700 guide the hammer 605 into alignment with the DTH tool 515 to avoid cross threading. FIG. 25D shows the lower and upper alignment clamps 700', 700 closed to maintain alignment as the hammer 605 is threaded onto the DTH tool 515.

As shown in FIG. 25E, once the joint is made, the lower and upper alignment clamps 700', 700 can be retracted to the open position, and as shown in FIG. 25F, the drill string 150, hammer 605, and DTH tool 515 are retracted out of the enclosure 195.

In various embodiments, the alignment clamps 700', 700 may be configured to be operated manually, electrically, pneumatically, hydraulically, or by other methods. In some embodiments, the alignment clamps 700', 700 may be operatively coupled to the control system 155 such that the rig operator, or an automatic process in the control system 155, determines when to move each of the alignment clamps 700', 700 from the open position to the closed position and vice versa.

None of the description in this application should be read as implying that any particular element, step, or function is an essential element which must be included in the claim scope; the scope of patented subject matter is defined only by the allowed claims. Moreover, none of these claims are intended to invoke 35 U.S.C. Section 112(f) unless the exact words "means for" are used, followed by a gerund. The

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claims as filed are intended to be as comprehensive as possible, and no subject matter is intentionally relinquished, dedicated, or abandoned.

The invention claimed is:

1. An apparatus for changing a drilling tool for a drilling rig, wherein the drilling rig comprises a drill tower supporting a drill pipe;
 - the apparatus for changing a drilling tool comprising:
 - a drilling tool changer assembly moveable between a storage position and an exchange position;
 - the drilling tool changer assembly supporting a rotatable carousel assembly;
 - the rotatable carousel assembly having a plurality of bit adaptors for holding drilling tools;
 - the rotatable carousel assembly selectively rotatable to bring a selected drilling tool into coaxial alignment with the drill pipe when the drilling tool changer assembly supporting the rotatable carousel assembly is moved into the exchange position; and
 - means for aligning the selected drilling tool within a bit adaptor holding the selected drilling tool.
2. The apparatus for changing a drilling tool of claim 1, further comprising:
 - means for aligning the drill pipe with the selected drilling tool.
3. The apparatus for changing a drilling tool of claim 1, wherein the selected drilling tool is a down-the-hole ("DTH") tool.
4. The apparatus for changing a drilling tool of claim 3, further comprising:
 - a hammer coupled to the drill pipe.
5. The apparatus for changing a drilling tool of claim 4, further comprising:
 - means for aligning the hammer with the DTH tool.
6. The apparatus for changing a drilling tool of claim 1, where the rotatable carousel assembly is further selectively rotatable to bring a selected bit adaptor of the plurality of bit adaptors into a pre-determined angular position for pick-up or drop-off of a drilling tool in the rotatable carousel assembly.
7. The apparatus for changing a drilling tool of claim 1, where the bit adaptors further comprise at least one nub for holding a drilling tool against a torque imparted to the drilling tool.

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8. The apparatus for changing a drilling tool of claim 1, where the drilling tool changer assembly is pivotably hinged to bring the rotatable carousel assembly into coaxial alignment with a drill pipe supported by the drill tower.

9. The apparatus for changing a drilling tool of claim 1, where the bit adaptors are differently sized to accommodate differently-sized drilling tools.

10. The apparatus for changing a drilling tool of claim 1, further comprising a control system for controlling operations of the drilling tool changer assembly.

11. The apparatus for changing a drilling tool of claim 10, further comprising a first actuator for selectively rotating the rotatable carousel assembly under command of the control system.

12. The apparatus for changing a drilling tool of claim 10, further comprising a second actuator for selectively moving the drilling tool changer assembly between the storage position and the exchange position under command of the control system.

13. The apparatus for changing a drilling tool of claim 1, where the drilling tool changer assembly further comprises a sensor for identifying the one or more drilling tools located in the bit adaptors.

14. The apparatus for changing a drilling tool of claim 1, where the drilling tool changer assembly further comprises a sensor for detecting an angular position of the rotatable carousel assembly.

15. The apparatus for changing a drilling tool of claim 1, wherein the drilling tool changer assembly further comprises a detector for detecting the presence or absence of a drilling tool in one of the plurality of bit adaptors.

16. The apparatus for changing a drilling tool of claim 1, where the drilling tool changer assembly further comprises an enclosure; the enclosure enclosing the rotatable carousel assembly.

17. The apparatus for changing a drilling tool of claim 16, where the drilling tool changer assembly further comprises a lid and a third actuator; the lid covering the enclosure; and, wherein the lid is deployed between an open position and a closed position by action of the third actuator.

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